INTEGRAL TOPOLOGY AND SMALL EIGENVALUES IN CALABI–YAU CONIFOLD TRANSITIONS

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Abstract. We determine the leading small spectrum of the Hodge Laplacian on real two-forms for Ricci-flat metrics in polarized Calabi–Yau conifold transitions. We assume a simply connected projective small resolution and a fixed smoothing that smooths every node to first order, approached along a fixed ray. Exactly as many positive eigenvalues tend to zero as there are divisor directions lost under smoothing, and their decay is inverse-logarithmic in the smoothing parameter. The leading coefficients are 2π times the generalized eigenvalues of the squared exceptional-period form relative to the harmonic norm after projection off all surviving cohomology. We compute this norm from global intersections. The divisor-period restriction map also determines the integral relations among cohomological vanishing cycles. The small eigenspaces are coexact and primitive of real type $(1,1)$, with a uniform gap in the remaining positive two-form spectrum. Applications to four-node and eight-node toric transitions give three and two distinct coefficients, respectively. For the sixteen-node split quintic with Kähler class $c_1(\mathcal{O}(1))$, the coefficient is 10π; we also calculate its integral rings, attachment and specialization maps, primitive vanishing lattice and monodromy image of index sixteen. The resolution and smoothing branches converge to a common Ricci-flat metric completion, and their tangent Hermitian–Einstein connections converge smoothly to the central tangent connection on compact subsets of the regular locus.

1. Introduction

We determine the leading small positive two-form eigenvalues of the Ricci-flat metrics in a polarized Calabi–Yau conifold transition. Their number is the loss of second cohomology under smoothing. Their coefficients depend on the periods of divisor classes over exceptional curves and on the global polarized intersection form. The same period map determines the integral relations among the vanishing cycles. Thus the topological and spectral conclusions use the same marked geometric transition. Local node types alone do not determine either the compact cohomology ring or these spectral coefficients.

We calculate the global data for the split quintic and for two toric transitions constructed in earlier work. The two toric applications use established geometry from the stability–degeneration series and from the separate Calabi–Yau smoothing series. We denote Papers IV–VI of the latter series by CY IV–VI below; these are distinct from the numbered stability–degeneration papers. The spectral theorem and its proof come first, followed by the compact applications and the precise roles of their geometric sources.

1.1. Geometric hypotheses and the main theorem.

Let X_0 be a normal projective threefold with trivial dualizing sheaf, with exactly $k \geq 1$ ordinary double points and no other

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singularities. Let

$$q : Y \longrightarrow X_0, \qquad f : \mathcal{X} \longrightarrow \Delta$$

be respectively a projective small crepant resolution and a proper flat polarized smoothing over a complex disc. Assume that Y is smooth and simply connected, that the relative canonical bundle of f is trivial, and that $X_t = f^{-1}(t)$ is smooth for $0 < |t| < t_0$. At each node the local smoothing function τ_i is required to satisfy

$$\tau_i(0) = 0, \qquad \tau_i'(0) \neq 0. \tag{1}$$

This is a condition on a genuine family over a disc. Let $C_i \simeq \mathbb{P}^1$ be the exceptional curves and impose

$$b_2(Y) = 2, \qquad b_2(X_t) = 1. \tag{2}$$

The polarization on X_0, its pullback to Y, and its extension to X_t are denoted by H. It is an integral class, with no implicit factor of 2π. Choose an integral divisor class D independent of H over $\mathbb{R}$, and put

$$v = H^3, \qquad a = DH^2, \qquad d = D^2H, \qquad b_i = D \cdot C_i, \qquad M = \frac{a^2}{v} - d. \tag{3}$$

All intersections in this display are taken on Y.

On X_t let g_t be the unique Ricci-flat Kähler metric with $[\omega_t] = H$. Let g_0 be the singular Ricci-flat Kähler metric in class H on $M_0 = X_0^{\mathrm{reg}}$, supplied by [EGZ09, HS17]. By central harmonic representatives we mean smooth L^2 two-forms on (M_0, g_0) that are closed and coclosed. Sections 3 and 6 construct the representatives used below and justify their domains and uniqueness. We use the nonnegative real Hodge Laplacian $\Delta_p = dd^* + d^*d$ on real p-forms and its ordinary L^2 norm. Write $\lambda_2^+(t)$ for its least strictly positive eigenvalue in degree two. The subscript indicates form degree. All limits below are along the fixed positive ray $0 < t < t_0$.

Theorem 1.1. *Under the preceding hypotheses, $M > 0$, and*

$$\lambda_2^+(t) = \frac{2\pi \sum_{i=1}^k b_i^2}{M\,|\log t|}(1 + o(1)). \tag{4}$$

For all sufficiently small t, this eigenvalue has real multiplicity one, and its real eigenspace is generated by a coexact primitive $(1,1)$-form. There is a constant $\gamma > 0$ such that the degree-one spectrum and every other positive degree-two eigenvalue are at least γ. Moreover,

$$\lambda_2^+(t) \leq \frac{2\pi \sum_i b_i^2}{M\,|\log t|} + C|\log t|^{-2}. \tag{5}$$

The constants refer to the fixed family, polarization, ray and local identifications. No uniformity as the leading coefficients in (1) degenerate is asserted.

Replacing D by $cD + eH$, with $c \neq 0$, multiplies both M and $\sum b_i^2$ by c^2. Thus the coefficient belongs to the one-dimensional divisor quotient modulo the surviving polarization. It is expressed in finite global intersection and restriction data; the list of local nodes alone does not determine it. The theorem gives a matching leading asymptotic and a sharper one-sided upper estimate. It does not assert a two-sided second-order remainder.

1.2. **The complete fixed-family cluster.** The same geometric problem allows several divisor directions to be lost. Retain the preceding family, polarization and simple-smoothing hypotheses, but replace (2) by $b_2(Y) = b$, $b_2(X_t) = s$ and $m = b - s > 0$. Set

$$V = H^2(Y;\mathbb{R}), \qquad r(D) = (D \cdot C_i)_i, \qquad K = \ker r, \qquad \mathcal{Q} = V/K.$$

The attachment sequence identifies K with the surviving cohomology $H^2(X_t;\mathbb{R})$. Define the forms

$$G(D,E) = -\int_Y DEH + \frac{3(\int_Y DH^2)(\int_Y EH^2)}{2\int_Y H^3}, \tag{6}$$
$$E([D],[E]) = r(D)\cdot r(E).$$

Let M be the quotient form induced by G on $\mathcal{Q}$, equivalently the minimum squared norm over each coset. The theorem includes the positivity needed for this definition. Write $\lambda^+_{2,j}(t)$ for the ordered positive real degree-two eigenvalues, with multiplicity.

Theorem 1.2. *The form G is the positive-definite inner product of central harmonic representatives, and E, M are positive definite on the m-dimensional space $\mathcal{Q}$. If $0 < \mu_1 \le \cdots \le \mu_m$ are the generalized eigenvalues of E relative to M, then*

$$\lambda^+_{2,j}(t) = \frac{2\pi\mu_j}{|\log t|}(1 + o(1)), \qquad 1 \le j \le m. \tag{7}$$

The degree-one spectrum and $\lambda^+_{2,m+1}(t)$ have a uniform positive lower bound. For small t, the entire m-dimensional positive small spectral space consists of coexact primitive real $(1,1)$-forms. A multiplicity-one μ_j gives an eventually simple eigenvalue. These assertions concern the fixed family and ray; no uniformity in degenerating node-scale ratios or two-sided next-order error is asserted.

The quotient removes all surviving harmonic directions, rather than only H. It is calculated by the intersection Schur complement (42). When $b = 2$ and $s = 1$, this gives the coefficient of Theorem 1.1; its one-sided sharper upper bound was proved separately. Section 6 supplies the full kernel projection and the ordered lower bounds needed for Theorem 1.2.

1.3. **Analytic inputs and proof.** Singular Ricci-flat metric existence, conical regularity and the local expansions for the actual smoothing metrics are supplied by [EGZ09, HS17, Ben26]. They do not determine the compact small-eigenvalue coefficients. We combine those metric results with conic spectral compactness and a sharp capacity calculation to prove (4) and the ordered matrix law (7).

Anné–Takahashi identify the spectral limits in exact-cone surgery in terms of the central operator, complete-cap harmonic forms and critical extended solutions [AT15]. Their work already contains logarithmic cutoff mechanisms and small coexact forms. The additional analytic conclusion here is the explicit coefficient in (4), including the matching lower bound for the actual Ricci-flat metrics. Its proof has three steps. First, the integral restriction map identifies the boundary periods of the lost divisor class; a scalar correction and vanishing boundary terms compute its primitive harmonic norm from cup products. Second, a sharp logarithmic capacity estimate on each Stenzel cap gives the local energy, and the actual tensor comparison controls the transported trial form and its codifferential. Third, conic spectral compactness excludes further small modes and mass loss; the same cap estimate gives the matching lower bound. A derivative-free comparison transfers that bound from exact-cone models to the actual metrics, with auxiliary radii sent to their limits after the family parameter.

For several lost directions, the harmonic norm must be quotiented by the entire surviving cohomology. We prove convergence of that harmonic space using closed compact approximants and uniqueness of harmonic representatives for the central metric. The lower bound is applied to every linear combination in a limiting spectral subspace; min–max then identifies each ordered coefficient. This supplies both the multiplicities and the matrix law.

1.4. **Integral topology and geometry on one compact transition.** Choose independent linear forms l_1, l_2 and quartics q_1, q_2 on $\mathbb{P}^4$. Put $\Pi = \{l_1 = l_2 = 0\}$ and assume that the restricted quartics meet transversely in sixteen points, that $X_0 = \{l_1q_2 - l_2q_1 = 0\}$ is smooth away from Π, and that

$$Y = \{ul_1 + vl_2 = 0,\ uq_1 + vq_2 = 0\} \subset \mathbb{P}^4 \times \mathbb{P}^1$$

is smooth. These are nonempty open conditions. Choose a quintic G nonzero at the sixteen nodes and let $X_t = \{l_1q_2 - l_2q_1 + tG = 0\}$, for sufficiently small t. The projective construction, the hyperplane markings H, L, and the chosen smoothing ray are the global inputs. The complement ring and attachment maps below are calculated, rather than assumed.

Let U be the common compact exterior, $q : Y \to X_0$ the projection, and $s_t : X_t \to X_0$ the continuous Milnor collapse along the ray. Mark the degree-two link generators by the exceptional rulings. Write

$$\Lambda = \mathbb{Z}^{16}, \qquad \mathbf{1} = (1, \dots, 1), \qquad \Lambda_0 = \ker(\Sigma : \Lambda \to \mathbb{Z}), \qquad W = \Lambda/\mathbb{Z}\mathbf{1}.$$

Here Σ is coordinate summation. Let J_{15} be the 15×15 all-ones matrix.

Theorem 1.3 (The marked split-quintic transition)**.** *The preceding construction has the following compatible properties.*

(1) *The smooth threefolds Y and X_t are simply connected and have torsion-free integral cohomology, with the full rings and Chern classes in Table 1. Their third Betti numbers are* 174 *and* 204*; their degree-three products are unimodular alternating pairings times the point class. In both cases $H^0 = H^6 = \mathbb{Z}$ and $H^1 = H^5 = 0$. The integral groups of U and X_0 are free, with Betti vectors*

$$(1, 0, 2, 189, 1, 15, 0), \qquad (1, 0, 1, 189, 2, 0, 1),$$

respectively. The central ring and both specialization pullbacks are given by (56) and the degree-three description in Section 7.4; in particular $\operatorname{im} s_t^* = \ker(T^* - I)$ in integral degree-three cohomology, where T is monodromy about a positive simple loop.

(2) *The degree-two boundary restriction is $r(hH + \ell L) = \ell\mathbf{1}$. It occurs in the integral exact sequence*

$$0 \longrightarrow \mathbb{Z}H \longrightarrow \mathbb{Z}H \oplus \mathbb{Z}L \xrightarrow{r} \Lambda \xrightarrow{\partial} H^3(X_t; \mathbb{Z}) \longrightarrow H^3(U; \mathbb{Z}) \longrightarrow 0. \tag{8}$$

The classes ∂e_i are the left Poincaré duals of the vanishing spheres. They generate the primitive isotropic lattice W, with the single relation $\sum_i \partial e_i = 0$. Under Poincaré duality the nonzero monodromy block is $\pm(I_{15} + J_{15})$, whose Smith form is $\operatorname{diag}(1^{14}, 16)$. Thus the monodromy image has index sixteen in the primitive vanishing lattice.

(3) *Use the actual Ricci-flat class $H + s^2L$ on Y and class H on X_t, where $s, t > 0$ are independent branch parameters. Both branches converge to the same metric completion (X_0, d_{g_0}), homeomorphic to X_0, with the local rescaled resolved-conifold and Stenzel limits. Every exceptional area on Y equals s^2. The same class L has period one at*

every link, and its central primitive harmonic representative α satisfies

$$[\alpha] = L - \tfrac{4}{5}H, \qquad \|\alpha\|^2_{L^2(g_0)} = \tfrac{16}{5}, \qquad \lim_{t\downarrow 0} |\log t|\lambda_2^+(t) = \frac{2\pi\|r(L)\|^2}{16/5} = 10\pi. \tag{9}$$

The eigenvalue has the simplicity, type, gap and one-sided upper-error properties of Theorem 1.1. The resolution and smoothing limits require no relation between s and t.

(4) *The tangent bundles on the smooth spaces are stable and their Ricci-flat tangent connections are Hermitian–Einstein. The relative tangent sheaf of the smoothing is flat, with central fibre E_0 and*

$$0 \longrightarrow E_0 \longrightarrow T_{X_0} \longrightarrow \bigoplus_{i=1}^{16} \mathbb{C}_{p_i} \longrightarrow 0, \qquad E_0^{**} = q_*T_Y = T_{X_0}, \qquad R^j q_* T_Y = 0 \ (j>0).$$

These extensions agree on the regular locus. Their actual tangent connections have the local limits and curvature-energy bounds in Proposition 8.2.

The common restriction map in (8) and (9) is essential. Its kernel gives the surviving divisor direction, its image gives the vanishing-cycle relation, and its periods give the local terms in the spectral energy. The polarized cup product supplies the global harmonic norm. Proposition 3.2 proves this compatibility for the general rank-one setting. Sections 7 and 8 calculate the compact example and complete the proof of Theorem 1.3. The topology asserted here is the integral cohomology and the specified maps; a classification of homotopy or diffeomorphism types is not asserted.

1.5. **An eight-node toric transition.** Let $B = \mathbb{P}^1 \times \mathbb{P}^1$ and $Z = \mathbb{P}_B(\mathcal{O} \oplus \mathcal{O}(1,-1)) \times_B \mathbb{P}_B(\mathcal{O} \oplus \mathcal{O}(1,-1))$. In the quotient convention write f, g for the base hyperplane classes and ξ_1, ξ_2 for the tautological classes. The anticanonical contraction of a general anticanonical hypersurface $Y \subset Z$ is the eight-node threefold X_0 constructed in [W26e, Appendix A, Proposition A.1].

Corollary 1.4. *This transition admits polarized smoothings with trivial relative canonical bundle and first-order smoothing at every node, whose central polarization pulls back to $H = (4g + 2\xi_1 + 2\xi_2)|_Y$. For every such fixed family and ray, the actual Ricci-flat metrics in class H have precisely two small positive degree-two eigenvalues, with*

$$\lambda^+_{2,1}(t) \sim \frac{\pi}{|\log t|}, \qquad \lambda^+_{2,2}(t) \sim \frac{5\pi}{4|\log t|}. \tag{10}$$

Both are eventually simple, coexact and primitive real $(1,1)$. The degree-one spectrum and all other positive degree-two eigenvalues have a uniform positive lower bound for that fixed family.

Here $b_2(Y) = 4$ and $b_2(X_t) = 2$. The period matrix and the quotient mass are calculated from the earlier proposition's divisor intersections and two exceptional-curve types. Section 9 verifies the additional polarized-family hypotheses and proves the corollary. This is an application of Theorem 1.2; it makes no new integral saturation assertion for the eight-node family.

1.6. **A four-node toric transition.** Let X_9' be the four-node partial resolution on the nonmirror side of [W26k, Section 11.1], with the chosen projective small resolution Y. Section 10 specifies the ample Cartier class $\mathcal{L}_0$ by the eleven-ray support vector (66). Write $H = c_1(q^*\mathcal{L}_0)$.

Corollary 1.5. *The threefold X_9' admits a relatively polarized smoothing carrying exactly $\mathcal{L}_0$, with trivial relative canonical bundle and first-order smoothing at every node. One has $b_2(Y) = 6$, $b_2(X_t) = 3$ and $H^3 = 8712$. For every such fixed family and ray, the actual Ricci-flat metrics in class H have precisely three small positive degree-two eigenvalues, with*

$$
\begin{aligned}
|\log t|\lambda^+_{2,1}(t) &\longrightarrow \frac{2\pi}{7},\\
|\log t|\lambda^+_{2,2}(t) &\longrightarrow \frac{\pi(375-3\sqrt{3065})}{628},\\
|\log t|\lambda^+_{2,3}(t) &\longrightarrow \frac{\pi(375+3\sqrt{3065})}{628}.
\end{aligned}
\tag{11}
$$

All three are eventually simple, coexact and primitive real $(1,1)$. The degree-one spectrum and all other positive degree-two eigenvalues have a uniform positive lower bound for that fixed family.

The source's integral exceptional relation $(1,-1,-1,1)$ supplies a full-support deformation tangent through [W26j, Theorems 2.5 and 4.5]. We prove that the exact ample class extends, and calculate the full three-dimensional harmonic quotient. The resulting spectral law is an additional consequence on the established original-side geometry; it is not a claim about the mixed-singularity X_9 or the distinct mirror family.

1.7. **Global inputs and the earlier reconstruction results.** The small-spectrum theorem for two-step bundles [W26h, Theorem 1.1] and the local Hermitian–Einstein reduction [W26f, Theorem 1.1] relate finite degeneration data to endomorphism Dolbeault spectra on a smooth base. The present operator is the real Hodge Laplacian on a degenerating threefold. Those fixed-background estimates are not premises of its spectral proof. The shared question is which global geometric data must be retained to determine the limiting analysis.

The local link and its two fillings do not determine the compact cohomology ring. They must be attached to a specified global exterior. In the K3 bundle-moduli setting, [W26d, Theorem 3.1 and Proposition 3.2] retains the marked complement and geometric overlap maps; its monodromy example explains why local fibre rings are insufficient. The explicit calculations in [W26b, W26c] supply such global information for particular compact contractions. Theorem 1.3 follows the same distinction: the projective family supplies the information from which its exterior ring and attachment maps are computed. For Theorem 1.1 alone, the intersection numbers and restriction vector remain global inputs.

The K3 results concern moduli spaces of bundles on a smooth surface; their harmonic metric is the L^2 metric on bundle deformations. The canonical singular Ricci-flat metrics in [W26a, Theorem 1.1] concern compact K3 moduli spaces that admit neither a symplectic resolution nor a smoothing. Here the compact base threefold admits both branches of a conifold transition, and its Ricci-flat metric defines the Hodge operator. We use projective complete-intersection topology, ordinary-node surgery, and conical Ricci-flat analysis in place of the K3 Mukai lattice and hyperkähler local moduli models. These K3 papers provide the reconstruction questions and precedents; the proofs below establish the conifold conclusions in the new setting.

The role of [W26e, Appendix A, Proposition A.1] is different: its compact construction and calculated exceptional data are used directly in Corollary 1.4. Theorem 1.2 extends the analytic result to the two lost directions of that family. The earlier work distinguishes tangent-bundle stability walls from toric birational walls and Calabi–Yau cycle relations; the spectral application preserves that distinction.

The spectral law is a forward determination from the marked geometric data. Its finite set of coefficients does not recover the whole intersection ring, complex structure or metric. Likewise the tangent connection, being induced by the Ricci-flat metric, requires no separate arbitrary- bundle existence theorem. It does not provide the general bundle or flow analysis of [W26f, W26g] on a collapsing background.

1.8. **Conifold topology and deformation theory.** The split-quintic transition and its cohomological relations are classical [GMS95, Ros06]; the relation between exceptional and vanishing cycles is also central to the conifold analysis of [LLW18]. Singular Ricci-flat metrics with bounded potentials and their conical asymptotics are supplied by Eyssidieux–Guedj–Zeriahi and Hein–Sun [EGZ09, HS17]. Benabida proves the metric-valued expansions for actual resolutions and smoothings used here [Ben26]. These existence and local metric results are inputs to our quantitative comparison.

CY IV proves a vanishing-cycle obstruction for smoothings with nodes and specified del Pezzo cone points, allowing arbitrary analytic smoothing arcs [W26i, Theorem 6.1]. CY V identifies the image of the global deformation tangent map and proves smoothness of the selected deformation bases, giving a criterion for the prescribed mixed singularities [W26j, Theorems 2.5, 4.5 and 4.7]. Their purely nodal specialization is the classical full-support exceptional-curve relation criterion. An arbitrary smoothing arc need not satisfy (1); the selected tangent and the extension of the specified polarization must be checked separately. Neither the mixed-singularity criterion nor its nodal specialization computes the harmonic norm or the small Hodge spectrum studied here.

The relevant two maps have different domains. Over $\mathbb{Q}$, set

$$A : \mathbb{Q}^k \longrightarrow H_2(Y;\mathbb{Q}), \quad e_i \longmapsto [C_i], \qquad r = A^*, \qquad \operatorname{im} r = (\ker A)^{\perp}. \tag{12}$$

Here $r(D) = (D \cdot C_i)_i$, and the annihilator uses the marked node coordinates. The space $\ker A$ supplies localized deformation tangents; $\ker r$ is the surviving second cohomology, and $\operatorname{im} r$ supplies the spectral periods. The divisor-pairing description already appears in [W26i, Proposition 8.3]. The transpose identity is elementary linear algebra. Our spectral conclusion additionally requires the specified polarization, all its mixed cubic intersections, and the harmonic quotient by the entire surviving space. A relation lattice alone determines none of those norms.

CY VI identifies the original-side exceptional-curve relation lattice with a vanishing-sphere relation lattice on the prescribed Batyrev mirror [W26k, Theorem A]. This is distinct from the same-transition attachment duality in Proposition 3.2, where the relations among the smoothing's vanishing classes are $\operatorname{im} r$, not $\ker A$. The integral exterior sequence in CY VI, Section 9.7, is a preceding use of the same classical surgery tools. Its mirror theorem is not a premise of the general spectral proof. Over the integers, actual attachments and torsion must still be retained; a rational annihilator identity does not establish saturation or the complete cohomology ring.

Spectral convergence and low-energy resolvent analysis in related conic geometries are developed in [Row08, GS15]. The latter's Theorem 1 excludes the present harmonic link two-form channel through its condition (4). For tangent bundles, Collins–Picard–Yau prove HYM

existence and local control on balanced conifold transitions [CPY24]. Our tangent result uses the Chern connections of the actual Ricci-flat Kähler metrics and separately identifies the flat central sheaf and its reflexive hull.

For the integrated result, integral Lefschetz and the projective cubic form first determine the smooth rings. Thom and Mayer–Vietoris sequences compute the exterior and its attachments. Naturality of the actual contraction and Milnor collapse then gives the specialization maps and identifies the primitive vanishing lattice and monodromy. The class used in these calculations is the same L used for exceptional areas and the primitive harmonic norm. Finally the relative Jacobian sequence identifies the tangent sheaf, while the actual metric limits give its geometric interpretation. These calculations make the global inputs to the new spectral coefficient explicit; classical topology and metric existence are not claimed as separate new theories.

Section 2 fixes the analytic normalization and actual metric comparison. Sections 3–5 prove the restriction, harmonic-mass and spectral assertions. Section 6 proves the complete fixed-family cluster law. Sections 7–8 prove the compact integrated result. Sections 9 and 10 apply the cluster law to established families in the two programmes. Section 11 records the remaining reconstruction and analytic questions.

2. Conifold geometry and actual metric comparison

2.1. **The link and its integral normalization.** An ordinary double point is analytically the cone $\sum_{j=1}^{4} z_j^2 = 0$. Its Ricci-flat metric cone is $g_C = dr^2 + r^2 h$, where the homogeneous metric on $T^{1,1} \simeq S^2 \times S^3$ is

$$h = \frac{1}{9}\eta_c^2 + \frac{1}{6}\sum_{j=1}^{2}(d\theta_j^2 + \sin^2\theta_j\, d\phi_j^2), \qquad \eta_c = d\psi + \cos\theta_1 d\phi_1 + \cos\theta_2 d\phi_2. \tag{13}$$

Here ψ has period 4π and $\mathrm{Ric}_h = 4h$. Put $\sigma_j = \sin\theta_j d\theta_j \wedge d\phi_j$ and

$$\vartheta = \frac{\sigma_1 - \sigma_2}{8\pi}. \tag{14}$$

The circle bundle over $\mathbb{P}^1 \times \mathbb{P}^1$ has Chern class $(1,1)$ up to a common sign. A primitive two-cycle projects with degrees $(1,-1)$; therefore ϑ has period one. It is harmonic: it is closed, its horizontal part is anti-self-dual, and wedging that part with $d\eta_c = -(\sigma_1 + \sigma_2)$ gives zero. In the orthonormal frame,

$$|\vartheta|_h^2 = \frac{9}{8\pi^2}, \qquad \mathrm{Vol}(T^{1,1}, h) = \frac{16\pi^3}{27}, \qquad c := \|\vartheta\|_{L^2(h)}^2 = \frac{2\pi}{3}. \tag{15}$$

The selected small-resolution ruling fixes the sign of its primitive period. On a cone, ϑ is a primitive real $(1,1)$-form, of pointwise size $O(r^{-2})$.

The smoothing cap is T^*S^3 with the Stenzel metric, asymptotic with positive differentiated rate to this cone [Ste93, CO90]. The small-resolution cap is the total space of $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ with its Ricci-flat resolved-conifold metric. The two caps have the same link but retain different primitive cohomology: restriction to the link is an isomorphism in degree three for the smoothing cap and in degree two for the resolution cap.

2.2. **The central metric and the smoothing scale.** Let X_0^{reg} carry the singular Ricci-flat Kähler form ω in class H. The bounded-potential construction [EGZ09, §7.2, Theorem 7.5 and Corollary 7.6] gives a closed positive pullback current on Y. Its Bedford–Taylor products have their cohomological masses and give no mass to the exceptional analytic set. The conical

theorem of Hein–Sun [HS17, Theorem 1.4 and Corollary 1.5] supplies holomorphic node charts such that, for some $\mu > 0$,

$$|\nabla_C^j(g_0 - g_C)|_{g_C} \le C_j r^{\mu-j} \quad (j \ge 0). \tag{16}$$

Use the minimum rate over the finitely many ends. Benabida's Theorem A provides the corresponding polyhomogeneous expansion [Ben26]. The cited metric theorems use the class $2\pi c_1(\mathcal{L})$ for the polarizing line bundle $\mathcal{L}$. We rescale their metrics by $(2\pi)^{-1}$ to obtain our class $H = c_1(\mathcal{L})$, with the corresponding fixed rescaling of cone radii and cap normalizations. Written as $dr^2 + r^2 h$, the normalized link metric and (15) are unchanged.

Condition (1) makes the total-space projection a holomorphic Morse function near each node. The local family therefore has the weighted ordinary-node model, with weights $3/2$ on the coordinates and 3 on the algebraic smoothing parameter. This model is compatible with the central conical chart. Indeed, the transition between two ordinary-quadric charts lifts to an ambient holomorphic germ Φ with $Q \circ \Phi = uQ$, where $Q = \sum z_j^2$ and u is a holomorphic unit. Its linear part A satisfies $Q(Az) = u(0)Q(z)$ and extends to the weighted smoothing after a fixed rescaling of the base. Since all coordinate weights are $3/2$, the terms $\Phi - A = O(|z|^2)$ have relative conic order $O(r^{3/2})$, with differentiated bounds. After this linear normalization, (16) therefore holds in the family chart, with μ decreased if necessary. Along the fixed positive ray we use the reference metric scale

$$\epsilon = t^{1/3}. \tag{17}$$

On the smoothing branch, metrics, Kähler forms and eigenvalues indexed by ϵ are evaluated at the algebraic parameter $t = \epsilon^3$. The individual cap scales have fixed positive leading ratios to ϵ. Multiplying a local normalization by a fixed constant changes a logarithmic length by a bounded term.

Proposition 2.1 (Actual metric comparison)**.** *There are collar-compatible exterior identifications F_ϵ and constants $a_0 > 0$, $r_0 > 0$ such that, for fixed sufficiently large A,*

$$|\nabla_C^j(F_\epsilon^* g_\epsilon - g_0)|_{g_C} \le C_j(\epsilon/r)^{a_0} r^{-j}, \quad A\epsilon \le r \le r_0, \quad j = 0, 1. \tag{18}$$

The same comparison holds for the Kähler forms. On fixed exterior compact sets the C^1 differences are $O(\epsilon^{a_0})$. On fixed compact subsets of each rescaled cap, the metrics converge in all derivatives to the normalized Stenzel metric.

Proof. We specify the consequence of Benabida's Theorem C used here, including its differentiable identification. The ordinary-node model verifies the positive weighted smoothing hypothesis S.1; the Stenzel Kähler form is exact outside a compact set, as required by S.2. The family is polarized with trivial relative canonical bundle, and the central metric satisfies (16). The finite-node modification is the one described in [Ben26, Remark 5.2]; the corners are disjoint and their leading scale ratios are fixed.

In the product corner of [Ben26, §5.3.2, Proposition 5.9], let ρ_1, ρ_2 be boundary functions with $\epsilon = \rho_1\rho_2$. Map the central point $(y, r, 0)$ to $(y, r, \epsilon/r)$ on the smoothing. This preserves the central collar radius $r = \rho_1$. Extend these maps across the fixed exterior by choosing a horizontal lift agreeing with the prescribed collar lifts, using a partition of unity away from the smaller collars. Its flow supplies a trivialization smooth to the central face. No holomorphicity of F_ϵ is asserted.

The vertical b-tangent frame is $\rho_1\partial_{\rho_1} - \rho_2\partial_{\rho_2}, \partial_{y_j}$. Dividing by ρ_1 gives, after pullback, the cone frame $\partial_r, r^{-1}\partial_{y_j}$. The final metric-valued expansion is the Ricci-flat conclusion of [Ben26, Theorem C], expressed in the vertical conic bundle of [Ben26, Definition 5.1].

The formal correction in [Ben26, Proposition 6.1] has positive tensor weights, and the final correction at the end of its §7 is flat at the boundary. Thus the corrected metric retains the prescribed face values; in particular its $\rho_2 = 0$ value is the full central metric g_0. Its nonnegative index sets and finite boundary value imply that the difference of coefficients is $O(\rho_2^{a_0})$ for some $a_0 > 0$, after decreasing the exponent to absorb positive-power logarithms. The assertion persists under b-differentiation, and an ordinary cone derivative costs r^{-1}. This gives (18). The Kähler-form and complex structure expansions give the corresponding form assertion. The fixed exterior estimate and rescaled-cap convergence are the respective face restrictions of the same expansion. □

It matters that (18) compares with the full central metric. A triangle inequality through g_C leaves an $O(r^\mu)$ term independent of ϵ, which is insufficient for the upper bound below. Nor does a positive-order expansion of a scalar potential alone establish this tensor assertion.

3. Harmonic representatives and intersection mass

3.1. **The two global cohomology directions.** Write $M_0 = X_0^{\mathrm{reg}} = Y \setminus \bigcup_i C_i$. Removing a submanifold of real codimension four does not change cohomology in degrees one and two. The Thom sequence consequently gives

$$H^1(M_0;\mathbb{R}) = 0, \qquad H^2(M_0;\mathbb{R}) = \mathbb{R}H \oplus \mathbb{R}D. \tag{19}$$

The common exterior has boundary a disjoint union of copies of $S^2 \times S^3$. In the smoothing each filling has zero second cohomology. Mayer–Vietoris therefore identifies the image of $H^2(X_t)$ in $H^2(M_0)$ with the kernel of restriction to these links. This image is exactly $\mathbb{R}H$, and the restriction vector of D is (b_i). In particular it is nonzero. In fact $b_i \neq 0$ for every i: otherwise $H \cdot C_i = D \cdot C_i = 0$ would make the effective curve C_i numerically zero against every divisor class on the projective Y.

The smooth fibres are simply connected. This follows from van Kampen, since replacing either filling by the other preserves the fundamental group: the fillings and links are simply connected. Both Y and X_t are compact, simply connected Kähler threefolds with trivial canonical bundle, and admit Ricci-flat metrics. They have full SU(3) holonomy in this setting. A nontrivial de Rham decomposition would have a one-complex-dimensional Ricci-flat factor, whose compact simply connected cover cannot exist; the irreducible hyperkähler alternative has even complex dimension. Thus $H^{2,0}(Y) = 0$, consistently with the divisor interpretation of all real second-cohomology directions.

Proposition 3.1. *There is a real closed $(1,1)$-form η on M_0 representing D, with $\Lambda_\omega \eta = 3a/v$, such that*

$$\alpha = \eta - \frac{a}{v}\omega \tag{20}$$

is closed, coclosed, primitive and in the Friedrichs form domain. For some $\nu > 0$ its expansion on end i is

$$\alpha = b_i \vartheta_i + O(r^{\nu-2}), \qquad \nabla_C^j(\alpha - b_i\vartheta_i) = O(r^{\nu-2-j}), \tag{21}$$

for every $j \geq 0$. Its norm and orthogonality are

$$\|\alpha\|^2_{L^2(g_0)} = \frac{a^2}{v} - d = M > 0, \qquad \langle \alpha, \omega \rangle_{L^2} = 0. \tag{22}$$

Proof. We use $d^c = \frac{i}{2}(\bar\partial - \partial)$, so $dd^c = i\partial\bar\partial$. Choose a smooth Hermitian metric on the line bundle with class D on Y, and let θ_b be its real normalized curvature form. A local small-resolution neighborhood retracts onto its exceptional $\mathbb{P}^1$. The node is rational, and the higher direct images of its structure sheaf vanish. On the inverse image of a sufficiently small Stein neighborhood, $H^1(\mathcal{O}) = H^2(\mathcal{O}) = 0$; equivalently one can use the infinitesimal-neighborhood filtration with quotients $\mathcal{O}(n)^{\oplus(n+1)}$. The exponential sequence identifies the local Picard group with $\mathbb{Z}$, by exceptional degree. Thus the line bundle is the b_ith power of the selected ruling bundle.

The normalized curvature $\sigma_1/(4\pi)$ of that ruling differs from (14) by $(\sigma_1+\sigma_2)/(8\pi)$, a constant multiple of $dd^c \log r$. Changing the local Hermitian metric therefore makes its curvature $b_i\vartheta_i$. The local logarithmic metric ratios u_i satisfy $u_i = O(1 + |\log r|)$ and $\nabla_C^j u_i = O(r^{-j})$ for $j \geq 1$. Patch them with disjoint cutoffs. Then

$$\theta = \theta_b + dd^c u$$

is a global real closed $(1,1)$-form representing D, equal to $b_i\vartheta_i$ near each end. The original smooth curvature has cone size $O(r^{-2})$ with differentiated bounds. Since the cone charts are holomorphic, type is preserved in this construction. Cone primitivity and (16) give $\Lambda_\omega\theta = O(r^{\mu-2})$.

The bounded-potential intersection products on Y give

$$\int_{M_0} \omega^3 = v, \qquad \int_{M_0} \theta_b\omega^2 = a, \qquad \int_{M_0} \theta_b^2\omega = d. \tag{23}$$

For the mixed products, pairing the closed current with smooth closed forms gives their cohomological values by Stokes. The products have no mass on the exceptional curves. Positivity of θ_b is unnecessary: write it as the difference of two smooth positive forms, using a large multiple of a fixed Kähler form, and apply linearity to the bounded- potential products [BT82, EGZ09].

Truncate every end at $r = \rho$. The transgression for replacing θ_b by θ in the second integral of (23) is $d^c u\wedge\omega^2$, whose boundary integral is $O(\rho^4)$. In the third integral it is $d^c u\wedge(\theta+\theta_b)\wedge\omega$, with boundary integral $O(\rho^2)$. Both vanish. The bulk terms are absolutely integrable, since their worst size is $O(r^{-4})$ against volume $O(r^5 dr)$. Consequently $\int \theta\omega^2 = a$ and $\int \theta^2\omega = d$.

Choose $0 < \nu < \min(\mu, 1)$ away from scalar indicial roots. The conical scalar Poisson isomorphism [Ben26, Proposition 3.4], with one cutoff constant for each end, solves

$$\Lambda_\omega dd^c f = \frac{3a}{v} - \Lambda_\omega\theta, \qquad \int_{M_0} f\, dV = 0. \tag{24}$$

Here the right side has zero integral because $\Lambda\theta\, dV = \theta\omega^2/2$ and $\mathrm{Vol}(M_0) = v/6$. It has weighted order $O(r^{\nu-2})$. The operator on the left is $-\Delta_0/2$ in our convention. The link Ricci bound implies that the first positive scalar homogeneous rate is at least one; below ν only constants occur. The solution therefore has $f = f_i + O(r^\nu)$, with differentiated bounds on each end. The finite-end version is the direct sum of these constant asymptotic terms.

Put $\eta = \theta + dd^c f$. It has the required constant contraction, and (20) is primitive and closed. The Kähler identity $*\alpha = -\alpha\wedge\omega$ implies coclosedness. The new transgressions are $d^c f\wedge\omega^2$ and $d^c f\wedge(\eta+\theta)\wedge\omega$, with boundary integrals $O(\rho^{\nu+4})$ and $O(\rho^{\nu+2})$. Hence

$$\int \eta\omega^2 = a, \qquad \int \eta^2\omega = d,$$

and the primitive star identity proves

$$\|\alpha\|^2 = -\int \left(\eta - \frac{a}{v}\omega\right)^2 \omega = -d + \frac{a^2}{v}.$$

Primitivity gives pointwise orthogonality to ω. The nonzero restriction vector implies $\alpha \neq 0$, so this norm is positive. The expansion (21) follows from the scalar estimate; the bounded term proportional to ω is absorbed by $\nu < 1$.

Finally, let χ_ρ vanish for $r \leq \rho^2$, equal one for $r \geq \rho$, and interpolate smoothly with $|\chi_\rho'| \leq C/(r|\log\rho|)$. The removed L^2 mass is $O(\rho^2)$. Since $d\alpha = d^*\alpha = 0$,

$$\|d(\chi_\rho\alpha)\|^2 + \|d^*(\chi_\rho\alpha)\|^2 \leq \frac{C}{|\log\rho|^2}\int_{\rho^2}^{\rho}\frac{dr}{r} \longrightarrow 0.$$

This approximates α in the closed Friedrichs quadratic-form norm by compactly supported forms and proves the domain assertion. In particular α is in the Friedrichs kernel. The same argument applies to ω. □

The norm in (22) is the quotient norm after removing the surviving Kähler direction. This identifies the global mass needed in the spectral comparison without leaving an unspecified analytic coefficient among the input data.

3.2. **The restriction map in topology and in the spectral law.** The attachment sequence below is classical conifold topology [Ros06, LLW18]; compare the integral exterior sequence in [W26k, Section 9.7]. We retain its marked integral maps to identify the same divisor periods in the spectral coefficient. The curve-class map used in the smoothing criteria of [W26i, W26j] is its dual, as in (12). No mirror identification is used here. Let U be a compact truncation of M_0, with marked boundary $\coprod_i \mathcal{L}_i$. Let F_i be the smoothing filling, a disk cotangent bundle of S^3, and write δ_i for its oriented zero section in X_t. We use integral cohomology until real coefficients are specified. A left Poincaré dual is characterized by $\int_{X_t} \mathrm{PD}_L(\delta_i) \wedge \zeta = \int_{\delta_i} \zeta$. The integral class itself is defined by the oriented Thom class, so this convention also applies when its global image is torsion.

Proposition 3.2 (Restriction and spectral coefficient)**.** *Under the geometric hypotheses of Theorem 1.2, the actual attachments give the integral exact sequence*

$$0 \longrightarrow H^2(X_t;\mathbb{Z}) \longrightarrow H^2(Y;\mathbb{Z}) \xrightarrow{r} \mathbb{Z}^k \xrightarrow{\partial} H^3(X_t;\mathbb{Z}) \longrightarrow H^3(U;\mathbb{Z}) \longrightarrow 0, \tag{25}$$
$$r(D) = (D\cdot C_i)_i, \qquad \partial e_i = \mathrm{PD}_L(\delta_i).$$

Thus $\operatorname{im} r$ *is exactly the group of integral relations between the cohomological vanishing classes. In the rank-one case* (2)*,* $\ker(r\otimes\mathbb{R}) = \mathbb{R}H$*. On its one-dimensional quotient* $\mathcal{Q} = H^2(Y;\mathbb{R})/\mathbb{R}H$*, put*

$$P_H D = D - \frac{DH^2}{H^3}H, \qquad m_H([D]) = -\int_Y (P_H D)^2 H.$$

The form m_H *is positive definite, and the induced restriction* $r : \mathcal{Q} \to \mathbb{R}^k$ *satisfies*

$$\lim_{t\downarrow 0} |\log t|\lambda_2^+(t) = \frac{2\pi\|r([D])\|^2}{m_H([D])} \quad \text{for every } [D] \neq 0. \tag{26}$$

Here the Euclidean norm uses the primitive integral link generators. The same restriction coordinates are the boundary coefficients of the central primitive harmonic form used in the spectral proof.

Proof. The integral Thom isomorphism gives $H^j(Y,U;\mathbb{Z}) = \bigoplus_i H^{j-4}(C_i;\mathbb{Z})$. Consequently $H^2(Y;\mathbb{Z}) \to H^2(U;\mathbb{Z})$ is an isomorphism. The resolution filling retracts onto C_i, and its restriction in degree two to $\mathcal{L}_i$ is an isomorphism. Thus its marked boundary coefficient is $D \cdot C_i$.

Apply Mayer–Vietoris to $X_t = U \cup \coprod_i F_i$, with difference $j_U^* - j_F^*$. Since $H^1(\mathcal{L}_i) = H^2(F_i) = 0$, its degree-two part has the initial terms in (25). In degree three the filling restriction is an integral isomorphism, so the next difference map is

$$H^3(U;\mathbb{Z}) \oplus \mathbb{Z}^k \longrightarrow \mathbb{Z}^k, \qquad (y,z) \longmapsto r_3(y) - z.$$

Its kernel is the graph of r_3, canonically isomorphic to $H^3(U;\mathbb{Z})$ by projection. This proves exactness through the final exterior restriction. For the connecting class at the ith link, naturality factors the map through

$$H^2(\mathcal{L}_i;\mathbb{Z}) \longrightarrow H^3(F_i,\mathcal{L}_i;\mathbb{Z}) \longrightarrow H^3(X_t;\mathbb{Z}).$$

The first arrow takes the marked fibre-sphere generator to the integral Thom class of the zero section. The second is extension by excision. Orienting δ_i by the chosen collar convention identifies its image with $\mathrm{PD}_L(\delta_i)$. This proves the relation assertion integrally, without discarding torsion.

For the remaining assertions impose (2). After passage to real coefficients, the image of $H^2(X_t)$ is one-dimensional and contains H, hence is $\mathbb{R}H$. The projection $P_H D$ is unchanged by adding a multiple of H. Proposition 3.1 represents it by α and proves

$$m_H([D]) = \|\alpha\|^2 = \frac{(DH^2)^2}{H^3} - D^2H > 0.$$

On every link section, $[\alpha|_{\mathcal{L}_i}] = (D \cdot C_i)[\vartheta_i]$. Since $*_h\vartheta_i$ is closed, an exact discrepancy does not contribute to its pairing. Therefore, with $c = 2\pi/3$,

$$c^{-1} \int_{\mathcal{L}_i} \alpha|_{\mathcal{L}_i} \wedge *_h\vartheta_i = D \cdot C_i. \tag{27}$$

This is exactly the scalar boundary projection in Lemma 4.1. The upper and lower bounds proving Theorem 1.1 use these coefficients, normalized by the same harmonic mass. Substitution gives (26). Scaling $[D]$ multiplies its numerator and denominator by the same square. □

The proposition identifies the topological data in the spectral theorem; the analytic conclusion still requires its full metric and compactness arguments. In the general integral sequence, neither $\mathbb{Z}^k/\operatorname{im} r$ nor $H^3(U;\mathbb{Z})$ has been assumed torsion-free. Primitivity of the vanishing lattice is a separate conclusion in the split-quintic calculation below.

4. Capacity and the actual upper bound

4.1. **A sharp cap estimate.** Let (Z,g) be an oriented complete six-manifold with one asymptotically conical end, with cone $dr^2 + r^2h$ and

$$|\nabla_C^j(g - g_C)|_{g_C} \le C_j r^{-a_1-j}, \qquad j = 0,1, \qquad a_1 > 0.$$

Let ϑ be a harmonic two-form on the link and write $c = \|\vartheta\|^2_{L^2(h)} > 0$. Assume that the class of $*_h\vartheta$ extends to a closed three-form on a compact truncation Z_A. For a smooth two-form u on Z_T, let u_T denote its tangential boundary trace and set

$$B_u(T) = c^{-1} \int_L u_T \wedge *_h\vartheta, \qquad \mathrm{Cap}(T) = \inf_{B_u(T)=1} \int_{Z_T} (|du|^2 + |d_g^* u|^2)\, dV_g.$$

The infimum is over all smooth competitors; no elliptic boundary condition or existence of a minimizer is assumed.

Lemma 4.1. *For this fixed cap,*

$$\mathrm{Cap}(T) = \frac{c}{\log T} + O((\log T)^{-2}). \tag{28}$$

More precisely, every competitor satisfies

$$\int_{Z_T} (|du|^2 + |d_g^* u|^2)\, dV_g \geq \frac{c\,|B_u(T)|^2}{\log T + C_A} \tag{29}$$

for a fixed C_A and sufficiently large T.

Proof. Choose a closed three-form Ψ on Z_A whose boundary value is exactly $*_h\vartheta$. To obtain the exact value from a cohomological extension, subtract the discrepancy by an exact collar correction. Stokes and Cauchy–Schwarz give

$$cB_u(A) = \int_{Z_A} du \wedge \Psi, \qquad E_A(u) \geq \kappa_A |B_u(A)|^2, \quad \kappa_A = \frac{c^2}{\|\Psi\|^2_{L^2(Z_A)}} > 0.$$

On the end write $u = u_T(r) + dr \wedge v(r)$ and $B(r) = B_u(r)$. The projection of the radial component of du onto $dr \wedge \vartheta$ is $B'(r) dr \wedge \vartheta$, because $d_L^* \vartheta = 0$. Hence

$$\int_{A<r<T} |du|^2_{g_C} dV_C \geq c \int_A^T r |B'(r)|^2 dr.$$

The factor r follows from $|dr \wedge \vartheta|^2 = r^{-4} |\vartheta|^2_h$ and the six-dimensional volume density. For large fixed A, the AC comparison gives $w(r) = 1 - Cr^{-a_1} \geq 1/2$ such that

$$E_T(u) \geq \kappa_A B(A)^2 + c \int_A^T r w(r) |B'(r)|^2 dr.$$

Applying Cauchy–Schwarz to $B(T) = B(A) + \int_A^T B'(r) dr$ gives

$$|B(T)|^2 \leq \left(\kappa_A^{-1} + c^{-1} \int_A^T \frac{dr}{r w(r)} \right) E_T(u).$$

The radial integral is $\log(T/A) + O_A(1)$, which proves (29).

For the upper estimate, choose a smooth f vanishing near A and agreeing with $\log(r/A)$ for $r \geq 2A$. Extend $u = f(r)\vartheta / \log(T/A)$ by zero across the core. Its boundary coefficient is one and its exterior-differential energy is $c/\log(T/A) + O_A(\log^{-2} T)$. The metric error has integrable radial weight r^{-1-a_1} after the logarithmic factor is removed. For the codifferential, the cone identities $d_C^* \vartheta = 0$ and $\iota_{\nabla_C f} \vartheta = 0$ imply

$$|d_g^* \vartheta| \leq C r^{-3-a_1}, \qquad |\iota_{\nabla_g f} \vartheta| \leq C |f'| r^{-2-a_1}.$$

The product rule consequently gives

$$\|d_g^* u\|^2 \leq \frac{C_A}{\log^2(T/A)} \left(1 + \int_{2A}^T (1 + \log(r/A))^2 r^{-1-2a_1} dr \right) = O_A(\log^{-2} T).$$

This proves the matching upper estimate. □

For the Stenzel cap the extension condition holds: a compact truncation is the disk cotangent bundle of S^3, and restriction $H^3(Z_A;\mathbb{R}) \to H^3(S^2\times S^3;\mathbb{R})$ is an isomorphism. With the primitive normalization (15), $c = 2\pi/3$. Under $g \mapsto \epsilon^2 g$ the energy of two-forms is unchanged. Indeed the du term scales by $\epsilon^{6-2\cdot 3} = 1$; the other term has the same factor since $d^*_{\epsilon^2 g} = \epsilon^{-2} d^*_g$. An outer physical radius R therefore gives capacity

$$\frac{2\pi}{3\log(R/\epsilon)} + O(\log^{-2}(R/\epsilon)). \tag{30}$$

4.2. **A trial form for the actual Ricci-flat metric.** We now use the global form α of Proposition 3.1. Put $L_\epsilon = \log(1/\epsilon)$. On end i choose a cutoff $\chi_{i,\epsilon}$ vanishing below $A_i\epsilon$, equal to one above a fixed small R_i, and logarithmic between these radii:

$$\chi_{i,\epsilon}(r) = \frac{\log(r/(A_i\epsilon))}{\log(R_i/(A_i\epsilon))}.$$

Round its endpoints smoothly on fixed multiplicative annuli, making it flat at the inner endpoint. The rounding changes energy by $O(L_\epsilon^{-2})$. Denote the denominator by $L_i = L_\epsilon + O(1)$. Transport $\chi_\epsilon\alpha$ using F_ϵ and extend by zero across the cores, obtaining a smooth real two-form u_ϵ on X_t.

Proposition 4.2. *For the actual Ricci-flat metrics,*

$$\lambda_2^+(\epsilon) \le \frac{c\sum_i b_i^2}{ML_\epsilon} + O(L_\epsilon^{-2}), \qquad c = 2\pi/3. \tag{31}$$

Proof. Pullback commutes with d, so du_ϵ is the transport of $d\chi_\epsilon\wedge\alpha$. On end i its leading energy is $cb_i^2\int r|\chi_i'|^2dr = cb_i^2/L_i + O(L_\epsilon^{-2})$. The errors in the form, central metric and actual metric carry respectively positive factors r^ν, r^μ and $(\epsilon/r)^{a_0}$. Their integrals against dr/r are bounded, and each has prefactor $O(L_\epsilon^{-2})$. Thus

$$\|du_\epsilon\|^2 = \frac{c\sum_i b_i^2}{L_\epsilon} + O(L_\epsilon^{-2}). \tag{32}$$

The actual codifferential requires a separate estimate. Proposition 2.1 and the differentiated $O(r^{-2})$ bound for α give

$$|(d^*_{g_\epsilon} - d^*_{g_0})\alpha| \le C(\epsilon/r)^{a_0} r^{-3}.$$

Since $d^*_{g_0}\alpha = 0$, the contribution multiplied by χ_i has squared norm at most

$$C\int_{A_i\epsilon}^{R_i} \chi_i^2(\epsilon/r)^{2a_0}\frac{dr}{r} \le \frac{C}{L_\epsilon^2}\int_{A_i}^{\infty}(1+\log(s/A_i))^2 s^{-1-2a_0}ds = O(L_\epsilon^{-2}).$$

The product-rule contraction obeys

$$|\iota_{\nabla_{g_\epsilon}\chi_i}\alpha| \le C|\chi_i'|\big(r^{\nu-2} + (\epsilon/r)^{a_0}r^{-2}\big),$$

because the radial cone contraction of ϑ_i vanishes. Its squared norm has bounded radial integrals $\int r^{2\nu-1}dr$ and $\int(\epsilon/r)^{2a_0}dr/r$, again with prefactor $O(L_\epsilon^{-2})$. On the fixed exterior the squared error is $O(\epsilon^{2a_0})$. Combining with (32),

$$q_\epsilon(u_\epsilon) := \|du_\epsilon\|^2 + \|d^*u_\epsilon\|^2 = \frac{c\sum_i b_i^2}{L_\epsilon} + O(L_\epsilon^{-2}). \tag{33}$$

The mass density of α is $O(r\,dr)$, and the removed inner mass is $O(\epsilon^2)$. On the remainder, $|1-\chi_i| \le C(1+|\log r|)/L_\epsilon$, which is integrable against that density. The actual metric error integrates to $O(\epsilon^{\min(a_0,2)}(1+|\log\epsilon|))$. Therefore

$$\|u_\epsilon\|^2 = M + O(L_\epsilon^{-1}).$$

The Kähler-form comparison and pointwise central primitivity similarly give $\langle u_\epsilon, \omega_\epsilon\rangle = O(L_\epsilon^{-1})$. Subtract its orthogonal projection onto ω_ϵ. Since $\|\omega_\epsilon\|^2 = v/2$, this changes the squared norm by $O(L_\epsilon^{-2})$ and leaves q_ϵ unchanged. It removes the whole harmonic kernel because $b_2(X_t) = 1$. The Rayleigh principle then proves (31). □

5. Spectral compactness and the matching lower bound

The upper bound uses only one trial direction. To prove that its coefficient is sharp, we must identify the limit of every low-energy two-form and exclude mass concentrated in the caps. We first make the metrics exactly conical, apply conic spectral compactness at fixed auxiliary radii, and then remove that modification by a multiplicative comparison of eigenvalues.

5.1. **Comparison without derivatives of the metrics.**

Lemma 5.1. *Let g, h be smooth metrics on a closed n-manifold and suppose $(1-\delta)g \le h \le (1+\delta)g$, with $0 < \delta < 1/2$. For every ordered positive Hodge eigenvalue in degree p,*

$$K(\delta)^{-1}\lambda^+_{p,j}(g) \le \lambda^+_{p,j}(h) \le K(\delta)\lambda^+_{p,j}(g), \qquad K(\delta) = 1 + O_n(\delta).$$

No derivatives of the metric comparison are required.

Proof. The squared L^2 norms of p-forms satisfy

$$a_p\|u\|_g^2 \le \|u\|_h^2 \le b_p\|u\|_g^2, \quad a_p = (1-\delta)^{n/2}(1+\delta)^{-p}, \quad b_p = (1+\delta)^{n/2}(1-\delta)^{-p}.$$

The distributional differential d and its graph domain are unchanged under equivalent norms. Set $Z^p = \ker d$ and give $\operatorname{Dom} d/Z^p$ the quotient norm $\|[u]\|^2_{g,q} = \inf_{z\in Z^p}\|u+z\|_g^2$. The same constants compare these quotient norms. The coexact positive p-eigenvalues are the squared singular values of d on this quotient, with min–max quotient $\|du\|^2_{p+1,g}/\|[u]\|^2_{g,q}$. Its comparison factors are a_{p+1}/b_p and b_{p+1}/a_p. The positive exact p-spectrum equals the coexact $(p-1)$-spectrum. The full positive spectrum is the ordered union of these two lists, with multiplicity; the same common multiplicative bounds hold for their ordered union. This proves the lemma and avoids comparing codifferentials pointwise. □

5.2. **Fixed conic models.** On the cap end choose the radial variable $1/\rho_2$ from the product corner in Section 2. In the notation of [Ben26, §5.3.2], its restriction is $\sqrt{1+\widetilde{r}^2}$, where $\widetilde{r}$ is the original cap radius. This changes that radius by a relative $O(\widetilde{r}^{-2})$ term, with differentiated bounds, and preserves the asymptotically conical estimates after decreasing their positive decay rate if needed. Fix a small central radius $R > 0$ and a large cap radius $A > 0$. Let g_0^R equal g_0 outside $r = 2R$, equal g_C inside $r = R$, and interpolate between them. Let g_{St}^A agree with the normalized Stenzel metric up to radius A, with g_C beyond $2A$, and interpolate on the intervening annulus. These are smooth positive metrics; they need not be Kähler. Their relative errors are respectively $O(R^\mu)$ and $O(A^{-a_1})$.

For $2A\epsilon < R$, glue the fixed central metric to the scaled metrics $\epsilon^2 g_{\mathrm{St}}^A$, using the actual link attachments. The overlap is an exact cone. Fixed local normalization factors are absorbed in the finitely many cap metrics. Denote the resulting smooth metric on the same differentiable fibre by $\bar{g}_\epsilon^{R,A}$.

Lemma 5.2. *For some $b > 0$,*

$$\limsup_{\epsilon \downarrow 0} \sup_{X_t} |g_\epsilon - \bar{g}_\epsilon^{R,A}|_{\bar{g}_\epsilon^{R,A}} \leq C(R^\mu + A^{-b}). \tag{34}$$

The family limit is taken with R, A fixed.

Proof. On $r \geq R$, the actual metric converges to g_0, and the central interpolation costs $O(R^\mu)$. On each fixed rescaled cap the actual metric converges to Stenzel, and the cap interpolation costs $O(A^{-a_1})$. In the intervening region $A\epsilon \leq r \leq R$, Proposition 2.1 and (16) give relative cone error $O(r^\mu) + O((\epsilon/r)^{a_0})$, bounded by $C(R^\mu + A^{-a_0})$. The corner coordinates agree at the two faces: $r = \rho_1$ and the cap radius is $r/\epsilon = 1/\rho_2$. Thus these estimates compare the same conic frames. Taking $b = \min(a_0, a_1)$ proves the assertion. The errors tending to zero with ϵ may depend on R, A, which is why the limits are ordered. □

5.3. **Central harmonic forms and their domains.**

Lemma 5.3. *For g_0^R, the smooth L^2 closed and coclosed one-forms vanish. The corresponding two-form space maps isomorphically to $H^2(M_0; \mathbb{R}) = \mathbb{R}H \oplus \mathbb{R}D$. Every such two-form lies in the minimal graph domain of $d + d^*$.*

Proof. On an exact conical end write a closed and coclosed two-form as $u = a(r) + dr \wedge b(r)$. Link Hodge decomposition gives

$$a(r) = \sum_j c_j(r)\vartheta_j + d_L v(r), \qquad d_L^* v = 0,$$

where v is orthogonal to harmonic one-forms. The closedness equations imply $c_j' = 0$ and $b - v'$ closed. Coclosedness then gives $b = v' + b_H$ with b_H harmonic, and

$$r^{-2}\Delta_L v - v'' - 3r^{-1}v' = 0, \qquad b_H' + 3r^{-1}b_H = 0.$$

The latter term is $r^{-3}B_H$; as a radial two-form it is not L^2 at zero unless zero. On a positive coexact one-form eigenspace of link eigenvalue λ, the radial exponents of v are

$$k_\pm = -1 \pm \sqrt{1 + \lambda}.$$

The negative root is not L^2 as a two-form. Therefore

$$u = \sum_j c_j \vartheta_j + dv_+, \quad |v_+| = O(r^{\kappa - 1}), \quad |dv_+| = O(r^{\kappa - 2}) \tag{35}$$

for some $\kappa > 0$, with differentiated estimates. Orthogonal link eigenfunction expansion and elliptic estimates on fixed-ratio annuli justify these bounds; higher positive roots decay faster on a smaller end. The harmonic link part admits the logarithmic cutoffs of Proposition 3.1, and the positive-root part admits ordinary fixed-ratio cutoffs with vanishing graph-norm error. This proves the minimal-domain assertion.

If u is globally exact, choose a smooth primitive w. Its harmonic link coefficients vanish. Since $H^1(T^{1,1}) = 0$, $w - v_+$ is locally exact. Subtracting differentials of cutoff potentials makes $w = v_+$ near all ends. This primitive is in L^2 and in the minimal domain of d: the squared cutoff error is $O(\rho^{2\kappa+2})$. The adjoint identity then gives $\|u\|^2 = \langle d_{\min} w, u\rangle = \langle w, d_{\min}^* u\rangle = 0$. Thus the map to ordinary cohomology is injective.

For surjectivity, use the closed minimal de Rham complex. Its reduced cohomology $\ker d_{\min}/\overline{\operatorname{im} d_{\min}}$ has harmonic representatives by Hilbert-space orthogonal decomposition; no closed-range assertion is needed for this reduced statement. The forms ω, η from Proposition 3.1 remain closed minimal-domain forms for g_0^R, by norm equivalence. Their reduced

classes are independent: pair them with compactly supported closed four-forms dual to H, D under $H^2(M_0) \times H^4_c(M_0)$. Those pairings are L^2-continuous and vanish on the closure of minimal exact forms. The resulting reduced harmonic representatives are smooth, closed and coclosed on the interior. Injectivity into the two-dimensional ordinary cohomology now gives the isomorphism.

For degree one, the cone calculation gives tangential harmonic link one-forms and differentials of positive-root harmonic functions; the radial solution $Cr^{-5}dr$ is not L^2. The positive scalar roots are $-2 + \sqrt{4+\lambda} > 0$. Since $H^1(M_0) = 0$, a harmonic one-form has a global primitive that is bounded up to a constant on each end. Its cutoff error has squared norm $O(\rho^4)$. The same adjoint argument makes the form zero. □

Let H_R, D_R be the harmonic representatives in the lemma and set

$$\alpha_R = D_R - \frac{\langle D_R, H_R\rangle}{\|H_R\|^2} H_R, \qquad M_R = \|\alpha_R\|^2. \tag{36}$$

Then

$$M_R \longrightarrow M \quad \text{as } R \downarrow 0. \tag{37}$$

Indeed g_0^R and g_0 differ relatively by $O(R^\mu)$. Their equivalent norms define the same closed minimal complex and reduced cohomology. The harmonic norm of a class is the infimum over its closed representatives modulo the closed exact subspace. Hence all these norms differ by factors $1 + O(R^\mu)$. Polarization on the fixed two-dimensional space gives convergence of the Gram matrices, and their Schur complements converge as well. The limiting harmonic representatives are ω, η from Proposition 3.1. On the ith exact cone the harmonic link coefficient of α_R is the constant b_i, since H has zero restriction there.

5.4. **Cap kernels and the critical degree.** We record the two facts needed to exclude cap mass in the spectral compactness theorem. The complete cap with metric g^A_{St} is a scattering manifold after $x = 1/r$. The scattering Hodge theorem [HHM04, Theorem 1A] identifies its L^2 harmonic forms below middle degree with $H^j_c(T^*S^3)$, above middle degree with $H^j(T^*S^3)$, and in middle degree with $\mathrm{im}(H^3_c \to H^3)$. All vanish. The first and last ordinary groups vanish by retraction onto S^3 and duality; the middle map is the Euler-class map of the cotangent bundle of S^3, which is zero because S^3 is parallelizable. This conclusion does not require the interpolated metric to be Kähler.

On a completed exact-cone cap, a *critical extended solution* is a smooth solution of $(d + d^*)u = 0$ with expansion $u = u_{\mathrm{crit}} + O(r^{-3-\kappa})$ at infinity for some $\kappa > 0$, with differentiated bounds. Here u_{crit} is homogeneous of pointwise order r^{-3}; a nonzero such term has logarithmically divergent squared L^2 norm. The critical coefficients of these global solutions, modulo L^2 solutions, form the trace space $I_{1/2}$ in [AT15, equation (8)]. This is the space used in its logarithmically normalized critical term. Throughout, physical degree means ordinary differential-form degree.

Lemma 5.4. *An actual critical extended solution of $d + d^*$ on a complete six-dimensional exact-cone-ended cap has its logarithmic L^2 leading term in physical degree three.*

Proof. Such a solution has $|u| = O(r^{-3})$ and an expansion in powers from the discrete tangential spectrum, with a positive gap after its critical term. First, $du = d^*u = 0$ separately, even if u has mixed degree. Choose a cutoff χ_T supported in $r < 2T$, equal one below T, with $|d\chi_T| \le C/T$. The shell mass is bounded, so

$$\|(d + d^*)(\chi_T u)\|^2 \le CT^{-2}.$$

For compactly supported forms on a boundaryless manifold this is $\|d(\chi_T u)\|^2 + \|d^*(\chi_T u)\|^2$. On every fixed compact set letting $T \to \infty$ proves the separate equations.

A homogeneous pure p-form of pointwise order r^{-3} has the form $u = r^{p-3}a + dr \wedge r^{p-4}b$. The leading closed/coclosed equations are

$$d_L a = 0, \quad (p-3)a = d_L b, \quad d_L^* b = 0, \quad d_L^* a + (p-3)b = 0.$$

If $p \neq 3$, they imply $\Delta_L b = -(p-3)^2 b$, hence $a = b = 0$. For $p = 3$ they state precisely that a and b are harmonic, yielding $a + (dr/r) \wedge b$. Thus every critical leading term lies in degree three. The argument applies to traces of actual complete-cap solutions, not to arbitrary formal endpoint eigenvectors of a boundary operator. The faster remainder is L^2 at infinity. After division by $\sqrt{|\log \epsilon|}$, its contribution to the critical term therefore tends to zero in L^2. □

5.5. **Counting the small eigenvalues.**

Proposition 5.5. *For each fixed R, A, the metrics $\bar{g}_\epsilon^{R,A}$ have a uniform positive degree-one gap and exactly one positive degree-two eigenvalue tending to zero. That eigenvalue is eventually simple and coexact; the remaining positive degree-two spectrum has a uniform gap. Its normalized real eigenforms converge, along subsequences and after choice of sign, strongly on the central part and smoothly on fixed compact subsets to $\alpha_R/\sqrt{M_R}$.*

Proof. We apply the compactness and norm decomposition in [AT15, Theorems A–B, Corollary 17, equations (24)–(26), Lemma 18]. We retain the central extension selected by the cap trace space W in [AT15, equation (7)]. The additional link gap in [AT15, §5.1, p. 1047] identifies the minimal and maximal Gauss–Bonnet domains and allows a cohomological description of the full kernel. We do not use that identification. Instead, Lemma 5.3 places every relevant pure-degree closed and coclosed limit in the minimal domain, which is contained in the selected domain for every W. Its geometric hypotheses are satisfied: the compact exterior is fixed and exactly conical near its boundary, each compact cap has a conical collar, and the caps are scaled by a common ϵ, with fixed normalizations. Several disjoint caps replace the boundary operator and cap spaces by finite direct sums. All the local estimates in the proof sum over those components; this introduces no independent vanishing parameter.

For clarity, the consequence used here is the following norm compactness statement. A normalized sequence of eigenforms with vanishing eigenvalues has a subsequential central limit and possible additional mass in complete-cap L^2 harmonic forms and critical extended cap solutions. On the central region the sequence, up to an L^2 error tending to zero, is the sum of a term converging strongly to the central limit, a complete-cap tail, and a logarithmically normalized critical term. The squared norm is the sum of the limiting squared norms of those terms; orthogonality of finite collections is preserved. The critical term is the radial extension of the trace of an actual critical cap solution, with normalization $|\log \epsilon|^{-1/2}$.

Apply this statement to pure p-form eigenvectors u_j, $p = 1, 2$, with $\lambda_j \to 0$. On the compact smooth manifolds,

$$\|du_j\|^2 + \|d^* u_j\|^2 = \lambda_j \longrightarrow 0.$$

The central limit is pure degree p, since it agrees with the local limit of u_j on every fixed compact subset. Testing the separate energy bounds gives separate closedness and coclosedness. It is therefore in the space of Lemma 5.3. This argument makes no assertion that an arbitrary mixed-degree kernel of a central Gauss–Bonnet extension is invariant under degree projection.

The complete-cap term vanishes by the L^2 calculation above. Write the remaining three-term decomposition on the central part as $u_j = \Phi_j + \widetilde{\Phi}_j + \overline{\Phi}_j + o_{L^2}(1)$. Here Φ_j converges strongly to the central limit, $\widetilde{\Phi}_j \to 0$ by the cap-kernel vanishing, and $\overline{\Phi}_j$ is the critical term. By Lemma 5.4 the latter has physical degree three, up to an error tending to zero. Projecting this identity onto degree three kills u_j and its central limit; hence $\|\overline{\Phi}_j\| \to 0$. The norm identity now gives central norm one, with no lost mass. This projection is applied to the actual three-term decomposition, not commuted through its tangential spectral projectors.

The central dimensions zero and two give the upper counts in degrees one and two. Conversely approximate an orthonormal basis of the two central harmonic two-forms by compactly supported pure two-forms in the minimal graph domain. Their energies tend to zero and their Gram matrix tends to the identity. For each fixed cutoff extend them by zero across the shrinking caps. The model agrees with the central metric on their supports when ϵ is small. Min–max, first in ϵ and then in the cutoff radius, gives at least two eigenvalues tending to zero, including the harmonic eigenvalue. Since $b_1(X_t) = 0$ and $b_2(X_t) = 1$, precisely one positive two-form eigenvalue tends to zero. Failure of either asserted complementary gap would produce too many orthonormal zero-limit forms by compactness. The same count implies eventual simplicity. The degree-one gap excludes an exact small two-form, by compact Hodge decomposition.

It remains to identify its limit. Since H restricts trivially to the links, it has a compactly supported closed representative γ_H on M_0. Extend this by zero across the caps. Its model harmonic representative has bounded norm by minimization, and a fixed nonzero compactly supported dual pairing bounds its norm below. The preceding compactness at eigenvalue zero gives convergence without mass loss to H_R; compactly supported dual pairings identify its class uniquely. The small eigenform is orthogonal to this harmonic direction, so its central limit is $\pm\alpha_R/\sqrt{M_R}$. Interior elliptic estimates upgrade convergence to smooth convergence on fixed compact subsets. This proves the last assertion. □

5.6. **Sharp lower bound and proof of the main theorem.** Fix R, A and a cone section $r_* = R/2$ on each end. Let $B_{i,\epsilon}$ be the harmonic boundary coefficient of a normalized small model eigenform on that section. Proposition 5.5 and (35) imply, along every convergent subsequence and up to common sign,

$$B_{i,\epsilon} \longrightarrow b_i/\sqrt{M_R}.$$

Inside the section the metric is a scaled fixed exact-cone-ended cap. Lemma 4.1 therefore gives energy at least

$$\frac{cB_{i,\epsilon}^2}{\log(r_*/\epsilon) + C_{i,A}}.$$

These regions are disjoint. Summing and using unit total norm yields

$$\liminf_{\epsilon\downarrow 0} \bar{\lambda}_2^+(\epsilon; R, A)\log(1/\epsilon) \geq \frac{c\sum_i b_i^2}{M_R}. \tag{38}$$

The sign has disappeared on squaring. The argument applies to every sequence by subsequence compactness, so it establishes the full lower limit. The constants $C_{i,A}$ need no uniform bound in A, since the family limit is taken first.

Lemmas 5.1 and 5.2 transfer (38) to the actual metrics with factor tending to one as $A \to \infty$ and $R \downarrow 0$. By (37),

$$\liminf_{\epsilon\downarrow 0} \lambda_2^+(\epsilon)\log(1/\epsilon) \geq \frac{c\sum_i b_i^2}{M}.$$

Proposition 4.2 is the matching upper limit. Substitution of $c = 2\pi/3$ and $\log(1/\epsilon) = |\log t|/3$ proves (4), and (31) proves (5).

For the gaps, fix a single admissible pair R, A with small metric comparison error. The model gaps transfer by a fixed positive factor. Thus there is no need for a model gap uniform under the auxiliary limits. Simplicity and coexactness follow as in Proposition 5.5. On the actual Kähler manifold, type and conjugation commute with the Laplacian. A nonzero $(2,0)$ component would supply two independent real eigenvectors, contradicting simplicity. The eigenform is therefore real $(1,1)$. Its contraction by Λ_ω is a scalar eigenfunction with the same positive eigenvalue; if nonzero, its differential contradicts the degree-one gap. The eigenform is primitive. This completes the proof of Theorem 1.1.

6. Several lost divisor directions

We now retain the geometric hypotheses of Section 1 but replace (2) by $b_2(Y) = b$ and $b_2(X_t) = s$, with $m = b - s > 0$. The local metric, scalar and capacity arguments above do not depend on these dimensions. The additional issue is global: the trial forms must be projected off the entire s-dimensional harmonic space, and a lower bound is needed on every subspace of the small spectral cluster.

6.1. **The full harmonic space and its quotient norm.** Put $V = H^2(Y;\mathbb{R})$ and let $r : V \to \mathbb{R}^k$ be the exceptional-period map. The Thom and Mayer–Vietoris arguments of Proposition 3.2 give

$$K := \ker r = \operatorname{im}\big(H^2(X_t;\mathbb{R}) \longrightarrow V\big), \qquad \dim K = s.$$

In particular $H \in K$. Write

$$B(D,E) = \int_Y DEH, \qquad a(D) = \int_Y DH^2, \qquad v = \int_Y H^3. \tag{39}$$

Simple connectivity transfers to the smoothing by van Kampen, as before. Both smooth spaces have $H^{2,0} = 0$, so integral second-cohomology classes are represented by line bundles.

Apply the curvature construction and scalar correction in the proof of Proposition 3.1 to an integral basis of V, and extend linearly. For $D \in V$ this gives a closed and coclosed real $(1,1)$-form η_D with class D, constant contraction $3a(D)/v$, and link coefficients $r_i(D)$. The scalar inverse is fixed and linear; its solvability condition uses $a(D)$. The local Picard calculation, transgression estimates and logarithmic domain cutoffs do not use the dimension of V. After taking the minimum of finitely many positive rates, the resulting estimates hold uniformly on bounded subsets of V. Polarizing (22) gives

$$G(D,E) := \langle \eta_D, \eta_E \rangle = -B(D,E) + \frac{3a(D)a(E)}{2v}. \tag{40}$$

Indeed the primitive parts have pairing $-B(D,E) + a(D)a(E)/v$, and $\|\omega\|^2 = v/2$. The independent classes of the η_D make G positive definite.

We will need uniqueness for limits on the actual central metric, as well as on the exact-cone modifications.

Lemma 6.1. *Let a smooth metric on M_0 be uniformly equivalent near each end to the six-dimensional cone metric. Every smooth globally exact L^2 two-form u has an L^2 primitive in* $\operatorname{Dom} d_{\min}$. *Consequently a smooth L^2 closed and formally coclosed two-form is determined by its ordinary de Rham class.*

Proof. On one end write $u = a(r) + dr \wedge b(r)$ in fixed link coordinates. Choose v_0 with $d_L v_0 = a(r_0)$ and set

$$\beta(r) = v_0 + \int_{r_0}^{r} b(x)\, dx.$$

Closedness gives $d\beta = u$. In the fixed link norm put $B_0(x) = \|b(x)\|_{L^2(L,h)}$. Cone equivalence then implies $\int_0^{r_0} x^3 B_0(x)^2 dx < \infty$. For $0 < r < \delta < r_0$, Cauchy–Schwarz gives

$$r\|\beta(r)\|_{L^2(L)} \le rC_\delta + 2^{-1/2} \left(\int_0^{\delta} x^3 B_0(x)^2 dx \right)^{1/2}.$$

First let $r \to 0$, then $\delta \to 0$. Thus $r\|\beta(r)\|_{L^2(L)} \to 0$. The one-form β is L^2, since its squared cone norm has radial weight r^3. If χ_ρ vanishes below ρ and equals one above 2ρ, with derivative bounded by C/ρ, then

$$\|d\chi_\rho \wedge \beta\|^2 \le C\rho^{-2} \int_{\rho}^{2\rho} r^3 \|\beta(r)\|^2_{L^2(L)} dr = o(1).$$

The L^2 tails of β and $d\beta = u$ also vanish.

Let w_0 be a global smooth primitive. On the end $w_0 - \beta$ is closed and hence exact, since $H^1(L) = 0$. Subtracting the differential of a cutoff potential patches w_0 to β near each tip without changing its differential. The resulting global w is L^2, and the preceding cutoff estimate proves $w \in \operatorname{Dom} d_{\min}$. If u is formally coclosed, testing on the compactly supported core and passing to its graph closure gives $u \in \operatorname{Dom} d^*_{\min}$ and $d^*_{\min} u = 0$. Hence $\|u\|^2 = \langle d_{\min} w, u\rangle = 0$. Apply this to the difference of two forms to obtain uniqueness. □

For the actual central metric, this lemma and the b constructed forms identify the smooth L^2 harmonic two-form space with V. For g_0^R, the proof of Lemma 5.3 applies to a basis of b classes: compactly supported dual four-forms give b independent reduced classes, while its local expansion gives injection into ordinary cohomology. Its degree-one kernel is still zero. Denote the resulting central harmonic Gram form by G_R. Minimization in the same closed reduced complex and norm equivalence give $G_R \to G$, exactly as in (37).

Give $\mathcal{Q} = V/K$ the quotient norm M induced by G. The G-orthogonal lift $D^\perp$ of $[D]$ is primitive, because $G(D, H) = a(D)/2$. Thus

$$M([D],[E]) = -B(D^\perp, E^\perp), \qquad D^\perp \perp_B K. \tag{41}$$

The restriction of B to K is nondegenerate: its decomposition $K = \mathbb{R}H \oplus (K \cap \ker a)$ has one positive direction and a negative-definite primitive part. In a basis of K followed by quotient lifts, the mass matrix is therefore

$$\mathsf{M} = -B_{DD} + B_{DK} B_{KK}^{-1} B_{KD}. \tag{42}$$

The period form $E([D],[E]) = r(D) \cdot r(E)$ is positive definite on $\mathcal{Q}$. For g_0^R let M_R be the quotient norm induced by G_R. Then $M_R \to M$, and the generalized eigenvalues $\mu_j(R)$ of (E, M_R) converge to those of (E, M). Subtracting the polarization component alone would not give this quotient mass when $s > 1$.

6.2. **The surviving harmonic directions.** For $k \in K$, the form η_k has zero harmonic link coefficient. It admits on each end a primitive β_k of cone size $O(r^{\nu-1})$, with differentiated bounds. Indeed, write $\eta_k = a(r) + dr \wedge b(r)$. Its tangential coefficients in fixed link coordinates are $O(r^\nu)$, and $\partial_r a = d_L b$. Thus $\beta_k(r) = \int_0^r b(x)dx$ is a primitive with the claimed estimate. The bounded Kähler component satisfies the same bound after decreasing ν.

Choose χ_ρ equal one near the tip and zero for $r \geq 2\rho$. Replacing η_k on each end by

$$\gamma_{k,\rho} = \eta_k - d(\chi_\rho \beta_k) \tag{43}$$

gives a compactly supported closed form representing k, converging in L^2 to η_k. The removed tail and cutoff correction have squared norm $O(\rho^{2\nu+2})$. For fixed g_0^R its exact-cone expansion gives the same construction.

Let $h_t(k)$ be the actual harmonic representative of the corresponding class on X_t. Transport (43) and extend it by zero. It remains closed in that class, so minimization gives

$$\limsup_{t\to 0} \|h_t(k)\|^2 \leq \|\gamma_{k,\rho}\|^2_{g_0}.$$

Letting $\rho \to 0$ bounds this by $G(k,k)$. Local elliptic estimates give a subsequential smooth formally harmonic limit h_0 on M_0, which is L^2 by Fatou. Pairing with compactly supported closed dual four-forms identifies its class as k. Lemma 6.1 applied to $h_0 - \eta_k$ therefore gives $h_0 = \eta_k$. Lower semicontinuity proves the reverse norm inequality. Consequently the norms converge, no mass remains in the shrinking ends, and the convergence holds for the full family. Polarization over a basis proves convergence of the whole harmonic Gram matrix on K. The same proof works for each fixed exact-cone model with G_R.

6.3. **Upper bounds on all small subspaces.** For $D \in K^{\perp_G}$ apply the logarithmic trial construction of Section 4 linearly to η_D. Write $T_\epsilon D$ for the transported form, $L_\epsilon = \log(1/\epsilon)$ and $c = 2\pi/3$. The estimates there, polarized on this fixed finite-dimensional space, give

$$q_\epsilon(T_\epsilon D) = \frac{cE(D,D)}{L_\epsilon} + O(L_\epsilon^{-2})\|D\|^2, \tag{44}$$

$$\langle T_\epsilon D, T_\epsilon E\rangle = M(D,E) + O(L_\epsilon^{-1})\|D\|\|E\|. \tag{45}$$

The finite-dimensional norm on the right is arbitrary and fixed. The same mass estimates give strong exterior convergence to η_D and uniformly vanishing L^2 tails on bounded sets of D.

Project orthogonally onto actual real $(1,1)$-forms. The Kähler Hodge Laplacian preserves bidegree, so this projection does not increase q_ϵ. It changes the L^2 norm by $o(1)$: on fixed exterior compacts the complex structures converge and η_D is of type $(1,1)$; the tail estimate controls the remaining region. Next subtract the full harmonic projection. Its norm is $o(1)$ by the preceding subsection and $G(D,K) = 0$, and its removal leaves the energy unchanged. The resulting real $(1,1)$-forms $v_\epsilon(D)$ are orthogonal to the harmonic kernel, with mass Gram form tending to M.

Let $\mu_1 \leq \cdots \leq \mu_m$ be the generalized eigenvalues of (E,M). Apply min–max to the image under v_ϵ of the first j generalized eigendirections. Its dimension is j for small ϵ, and (44) gives

$$\limsup_{\epsilon\to 0} L_\epsilon \lambda^+_{2,j}(\epsilon) \leq c\mu_j, \qquad 1 \leq j \leq m. \tag{46}$$

The same bound holds within real $(1,1)$-forms. Only an $o(1)$ estimate was needed for the full kernel projection; this argument does not claim the stronger upper remainder of the rank-one theorem.

6.4. **Counting and the ordered lower bounds.** Fix the auxiliary radii R, A in the exact-cone model. The cap-kernel vanishing and physical-degree argument in Proposition 5.5 remain unchanged. The norm decomposition preserves every finite orthonormal collection of zero-limit one- or two-form eigenvectors as a central collection, without cap or critical mass.

The central dimensions are now zero and b. Conversely graph-norm approximation of a basis of all b central harmonic two-forms gives b zero-limit eigenvalues by min–max. Exactly s of these are the topological zeros. Hence there are precisely m positive small eigenvalues, and the complementary degree-one and positive degree-two gaps hold. Their orthonormal central limits belong to $K^{\perp_{G_R}}$, because the model's full harmonic kernel converges to K by the preceding argument.

Write $\bar{\lambda}_j(\epsilon; R, A)$ for these ordered positive model eigenvalues. Choose a sequence on which $L_\epsilon \bar{\lambda}_j$ has finite lower limit ℓ; an infinite lower limit requires no proof. The first j orthonormal eigenforms have a subsequence with orthonormal central limits $v_1, \dots, v_j \in K^{\perp_{G_R}}$. Every fixed linear combination $u_\epsilon(z)$ satisfies

$$q(u_\epsilon(z)) \le \bar{\lambda}_j |z|^2, \qquad v(z) = \sum_{a=1}^{j} z_a v_a.$$

At the fixed sections $r_* = R/2$, its harmonic boundary coefficients converge to $r_i(v(z))$. Apply Lemma 4.1 to this combination on every disjoint cap region and sum. It follows that

$$cE(v(z), v(z)) \le \ell |z|^2 = \ell M_R(v(z), v(z)).$$

One subsequence suffices for the finite basis, hence for every fixed combination; equivalently first use a countable dense set of z and then continuity. The largest generalized Rayleigh quotient on this j-dimensional span is at least $\mu_j(R)$. Therefore

$$\liminf_{\epsilon \to 0} L_\epsilon \bar{\lambda}_j(\epsilon; R, A) \ge c\mu_j(R). \tag{47}$$

The derivative-free comparison transfers this inequality to the actual metrics. Taking the family limit first, then $A \to \infty$ and $R \to 0$, removes its multiplicative error and sends $\mu_j(R)$ to μ_j. Together with (46) and $L_\epsilon = |\log t|/3$, this proves the law in Theorem 1.2. One fixed admissible pair R, A transfers the complementary positive gaps.

Finally, (46) supplies m small eigenvalues in real $(1,1)$-forms, and the total count is exactly m. Thus this type exhausts the small cluster. The degree-one gap excludes an exact small two-form. Contraction by Λ_ω commutes with the Kähler Laplacian; a nonzero contraction would give a positive scalar eigenfunction whose differential contradicts that same gap. Hence every small eigenform is primitive and coexact. A simple generalized eigenvalue μ_j is isolated from its neighbours, so the leading law and total count imply eventual simplicity. This completes the proof.

7. The split-quintic transition

We calculate the global data for Theorem 1.3, including the integral information that is not visible in the Hodge-number change. The geometric transition is classical [GMS95]; the following argument fixes the markings and attachment conventions used in this application. The calculation follows the distinction made for compact K3 bundle moduli in [W26d, W26c]: local fillings must be combined with global complement and attachment data. Here integral Lefschetz, the projective intersection form and the Thom sequences determine those data from (48).

7.1. **The genuine global family.** Choose independent linear forms l_1, l_2 and general quartics q_1, q_2 on $\mathbb{P}^4$, and set

$$\begin{aligned} F_0 &= l_1 q_2 - l_2 q_1, \qquad X_0 = \{F_0 = 0\}, \\ Y &= \{u l_1 + v l_2 = 0,\ u q_1 + v q_2 = 0\} \subset \mathbb{P}^4 \times \mathbb{P}^1, \\ X_t &= \{F_0 + tG = 0\}. \end{aligned} \tag{48}$$

Here G is a quintic nonzero at the sixteen points of $q_1 = q_2 = 0$ in $\Pi = \{l_1 = l_2 = 0\}$. The precise open conditions are transversality of the two restricted quartics, smoothness of X_0 off Π, and smoothness of Y.

Proposition 7.1. *These conditions hold on a nonempty open set. The projection $q : Y \to X_0$ is a projective small crepant resolution of exactly sixteen ordinary nodes. After shrinking the parameter disc, (48) is a flat polarized family with trivial relative canonical bundle and all noncentral fibres smooth. Every node is smoothed to first order. Both Y and X_t are simply connected.*

Proof. The linear system $l_1 q_2 - l_2 q_1$ is the full degree-five ideal of Π and is basepoint free off Π, so Bertini gives smoothness there. On Π its differential is $q_2 dl_1 - q_1 dl_2$. General plane quartics meet transversely in sixteen points by Bézout. At each intersection, (l_1, l_2, q_1, q_2) are analytic coordinates: the first two are normal to Π and the last two are independent along it. The local equation is the quadratic determinant, an ordinary double point.

The incidence divisor $ul_1 + vl_2 = 0$ is smooth, including over Π, where its normal differential $u\,dl_1 + v\,dl_2$ is nonzero. The second sections $uq_1 + vq_2$ form a basepoint-free system there. Bertini gives smooth Y. Projection has a unique kernel line at rank-one matrices and a whole $\mathbb{P}^1$ at the sixteen rank-zero matrices. The determinant coordinates identify its local resolution with $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. Adjunction for the two ample divisor classes $H + L$ and $4H + L$ gives $K_Y = 0$.

The relative hypersurface is flat over a sufficiently small disc. The condition $G(p_i) \neq 0$ gives a simple local smoothing; equivalently the holomorphic Morse lemma applies to F_0/G. Outside the node neighborhoods, compactness prevents a sequence of critical points from approaching the smooth central exterior. Thus all sufficiently small nonzero fibres are smooth. Adjunction gives trivial relative canonical bundle. Ehresmann's theorem with transverse boundary identifies the exteriors. Finally integral Lefschetz, twice for Y and once for the smooth quintic, gives simple connectivity and the integral degree-two markings used below. □

7.2. **The full integral rings.** Let H, L be the hyperplane classes of $\mathbb{P}^4, \mathbb{P}^1$ restricted to Y. Lefschetz gives $H_2(Y;\mathbb{Z}) = \mathbb{Z}^2$ and $H_2(X_t;\mathbb{Z}) = \mathbb{Z}$. Together with $H_1 = 0$, universal coefficients and integral Poincaré duality imply torsion-freeness in every degree. The fundamental class of Y in the product is $(H + L)(4H + L)$, whence

$$\int_Y H^3 = 5, \qquad \int_Y H^2 L = 4, \qquad L^2 = 0. \tag{49}$$

Take $A, B \in H^4(Y;\mathbb{Z})$ dual to H, L, and let $a_4 \in H^4(X_t;\mathbb{Z})$ satisfy $\int_{X_t} H a_4 = 1$. Then:

To verify the characteristic numbers directly, use

$$c(TY) = \frac{(1+H)^5(1+L)^2}{(1+H+L)(1+4H+L)}.$$

TABLE 1. Integral cohomology and characteristic data of the compact fibres.

	Y	X_t
H^2	$\mathbb{Z}H \oplus \mathbb{Z}L$	$\mathbb{Z}H$
H^4	$\mathbb{Z}A \oplus \mathbb{Z}B$	$\mathbb{Z}a_4$
Degree-two products	$H^2 = 5A + 4B$, $HL = 4A$, $L^2 = 0$	$H^2 = 5a_4$
Top products	$HA = LB = [\mathrm{pt}]$, $HB = LA = 0$	$Ha_4 = [\mathrm{pt}]$
rank H^3	174	204
c_2	$50A + 24B$	$50a_4$
Euler characteristic	-168	-200

This gives $c_2(Y) = 6H^2 + 5HL$ and $c_3(Y) = -20H^3 - 17H^2L$. For a quintic, $c_2 = 10H^2$ and $c_3 = -40H^3$. Since $H^1 = H^5 = 0$, the Euler numbers give the displayed third Betti numbers. The degree-three product on each compact fibre is its unimodular alternating pairing times the point class. With $H^0 = H^6 = \mathbb{Z}$, these statements specify the full rings up to symplectic basis choices. The product subgroup generated by H^2 and HL in $H^4(Y)$ has index $|\det \left(\begin{smallmatrix} 5 & 4 \\ 4 & 0 \end{smallmatrix}\right)| = 16$; it is not an integral basis.

Every exceptional curve has degrees $(H \cdot C_i, L \cdot C_i) = (0, 1)$. They therefore represent the same primitive homology class c_Y, with Poincaré dual B. Primitivity follows from the period of L.

7.3. **The common exterior and all cohomological attachments.** Remove disjoint tubular neighborhoods of the exceptional curves, obtaining a compact exterior U. Its boundary components $\mathcal{L}_i$ are $S^2 \times S^3$. Define

$$\Lambda = \mathbb{Z}^{16}, \quad \mathbf{1} = (1, \ldots, 1), \quad \Lambda_0 = \ker(\Sigma : \Lambda \to \mathbb{Z}), \quad W = \Lambda/\mathbb{Z}\mathbf{1}.$$

The homological Thom sequence uses $H_j(Y, U) = H_{j-4}(\coprod C_i)$. The maps $H_4(Y) \to \Lambda$ and $H_6(Y) \to \Lambda$ have primitive diagonal image: the former is exceptional intersection, witnessed

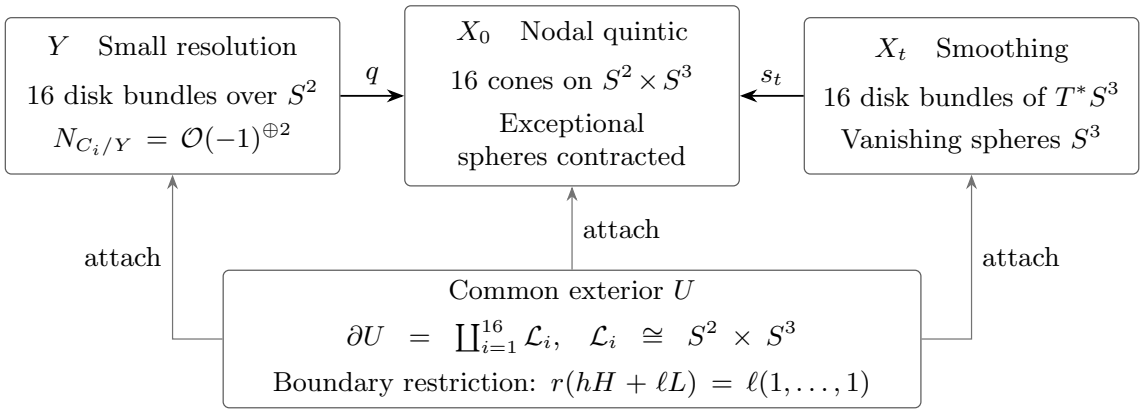


FIGURE 1. The common exterior and its three fillings. The map q is algebraic; s_t is the continuous Milnor collapse along the chosen ray. The resolution and smoothing cores have real dimensions two and three. The same boundary restriction gives the vanishing-cycle relation and the periods in the spectral coefficient.

by $L \cdot C_i = 1$, and the latter is the oriented diagonal fundamental class. Thus U has free integral homology and cohomology, with Betti vector

$$(1, 0, 2, 189, 1, 15, 0), \tag{50}$$

and exact sequences

$$0 \to W \to H_3(U) \to H_3(Y) \to 0, \qquad 0 \to H_3(U) \to H_3(X_t) \to \Lambda_0 \to 0. \tag{51}$$

The second comes from the Milnor fillings, whose cores are three-spheres. The vanishing spheres span W, with relation $\delta_1 + \cdots + \delta_{16} = 0$. Both successive quotients in (51) are free; hence W embeds primitively in $H_3(X_t)$, and is isotropic because the spheres are disjoint.

In cohomology the Thom sequence kills exactly $\mathbb{Z}B$ in degree four. Writing $\bar{A} = A|_U$, the even products are

$$H^2 = 5\bar{A}, \qquad HL = 4\bar{A}, \qquad L^2 = 0.$$

Orient each link as a boundary component of U and choose generators $\alpha_i \in H^2(\mathcal{L}_i)$ and $\beta_i \in H^3(\mathcal{L}_i)$ with $L|_{\mathcal{L}_i} = \alpha_i$ and $\int \alpha_i\beta_i = 1$. The symbol β_i here has degree three; the harmonic degree-two form remains ϑ_i. The degree-three restriction is

$$0 \to H^3(Y) \to H^3(U) \xrightarrow{r_U} \Lambda_0 \to 0, \tag{52}$$

and degree-five restriction is the inclusion $H^5(U) = \Lambda_0 \hookrightarrow \Lambda$. Boundary multiplicativity and that injectivity imply, for $y \in H^3(U)$,

$$Hy = 0, \qquad Ly = r_U(y) \in \Lambda_0 = H^5(U).$$

All degree-six products vanish. These formulas give the entire exterior ring, without supplying an unspecified multiplication as additional input.

For completeness, use the Mayer–Vietoris difference $j_U^* - j_F^*$. Let x_i be the degree-two generator of the resolution filling and z_i the degree-three generator of the smoothing filling, restricting to α_i, β_i. The nonzero difference maps are:

Degree	Resolution	Smoothing
0	$c_0\mathbf{1} - (c_i)$	$c_0\mathbf{1} - (c_i)$
2	$\ell\mathbf{1} - (x_i)$	$\ell\mathbf{1}$
3	$r_U(y)$	$r_U(y) - (z_i)$
5	$\Lambda_0 \hookrightarrow \Lambda$	$\Lambda_0 \hookrightarrow \Lambda$

Here $hH + \ell L$ is the exterior degree-two class, c_0 is an exterior constant, and (c_i) are filling constants. The nonzero connecting maps are

$$\begin{aligned} \partial_Y^3(v) &= (\Sigma v)B, & \partial_Y^5(v) &= (\Sigma v)[\mathrm{pt}], \\ \partial_{X_t}^2 : \Lambda \twoheadrightarrow W \hookrightarrow H^3(X_t), & & \partial_{X_t}^5(v) &= (\Sigma v)[\mathrm{pt}]. \end{aligned} \tag{53}$$

All other connecting maps vanish. The signs follow by representing the connecting class on an outward-U collar by $d\chi_F \wedge \eta$, where χ_F increases toward the filling. Pairing the degree-three connecting class of β_i with H, L gives zero and one respectively, hence $+B$. The same collar integral gives the top-degree sum maps.

Restriction from Y preserves H, L, sends (A, B) to $(\bar{A}, 0)$, and identifies $H^3(Y)$ with $\ker r_U$. Restriction from X_t sends (H, a_4) to $(H, \bar{A})$ and surjects $H^3(X_t) \to H^3(U)$ with kernel the cohomological copy of W in (53). Restriction to each Milnor filling is the corresponding component of $r_U(y)$. These data specify the full integral cohomological attachment diagram; no homotopy classification of the attaching maps is asserted.

7.4. **Monodromy and specialization.** Use left Poincaré duals in odd degree, characterized by $\int \mathrm{PD}_L(\delta)\wedge\eta=\int_\delta\eta$, and orient δ_i so that $\mathrm{PD}_L(\delta_i)=\partial^2_{X_t}(\alpha_i)$. For a positive simple loop, the Picard–Lefschetz transformation has $N=T-I$, $N^2=0$, and, up to the common intersection-sign convention,

$$N\gamma=\pm\sum_{i=1}^{16}\langle\gamma,\delta_i\rangle\delta_i. \tag{54}$$

Equivalently it factors through $H_3(X_t)\to\Lambda_0\hookrightarrow\Lambda\twoheadrightarrow W$. In the bases e_i-e_{16} of Λ_0 and $\delta_1,\dots,\delta_{15}$ of W, the nonzero block is $\pm(I_{15}+J_{15})$, where J_{15} is the all-ones matrix. Its Smith form is $\mathrm{diag}(1^{14},16)$: subtracting the first row from the others gives fourteen unit pivots, and the remaining determinant is sixteen. Thus the monodromy image has index sixteen in the primitive vanishing lattice. This relation-matrix description is consistent with the general conifold monodromy calculation of [LLW18].

The contraction $q:Y\to X_0$ collapses each exceptional sphere to its node. Along the chosen smoothing ray, the Milnor collapse $s_t:X_t\to X_0$ collapses each vanishing sphere. A disk cotangent bundle of S^3 modulo its zero section is the cone on its sphere bundle; the local collapses glue to the common exterior identification. Both maps have degree one. Collapsing the sixteen cores to their distinct nodes identifies the corresponding relative pairs with $(X_0,\{p_i\})$. In degrees at least two its relative groups are $H_j(X_0)$. The pair sequences give

$$\begin{aligned}&0\to H_3(Y)\xrightarrow{q_*}H_3(X_0)\to\Lambda_0\to 0,\\ &0\to W\to H_3(X_t)\xrightarrow{s_{t*}}H_3(X_0)\to 0.\end{aligned} \tag{55}$$

In degree two, q_* is the quotient by $\mathbb{Z}c_Y$ and s_{t*} is an isomorphism. In degree four, q_* is an isomorphism and s_{t*} is a primitive injection with free rank-one cokernel. To see the latter, the full relative sequence is

$$0\to H_4(X_t)\to H_4(X_0)\to\Lambda\xrightarrow{e_i\mapsto\delta_i}H_3(X_t)\to H_3(X_0)\to 0,$$

and the middle kernel is $\mathbb{Z}\mathbf{1}$. All central groups are free, with Betti vector $(1,0,1,189,2,0,1)$. The maps are isomorphisms in degrees zero and six; degrees one and five vanish.

The first sequence of (51) maps to the second sequence of (55), with identity on W, exterior inclusion in the middle, and q_* on the right. Both vertical cokernels are canonically Λ_0. This is naturality of the actual collapse maps, rather than an arbitrary splitting of free groups.

Let $h\in H^2(X_0)$ be the central polarization and choose $A_0,B_0\in H^4(X_0)$ by $q^*(h,A_0,B_0)=(H,A,B)$. The central even products and smoothing pullback are

$$h^2=5A_0+4B_0,\quad hA_0=[\mathrm{pt}],\quad hB_0=0,\qquad s_t^*(h,A_0,B_0)=(H,a_4,0). \tag{56}$$

Indeed pairing the degree-four pullbacks with H, and using degree one, fixes their coefficients integrally. Dually the image of s_{t*} in degree four is $\mathbb{Z}q_*\,\mathrm{PD}_Y(H)$. In degree three q^* is surjective with primitive kernel $\Lambda_0^*\simeq W$, and s_t^* is a primitive injection onto the annihilator of the vanishing homology lattice. Since $\operatorname{im}N$ has finite index in W, an integer-valued homomorphism annihilates $\operatorname{im}N$ exactly when it annihilates W. Thus

$$\operatorname{im}s_t^*=\ker(T^*-I)\subset H^3(X_t;\mathbb{Z}).$$

The central degree-three cup pairing factors through q^* and the unimodular pairing of Y, by naturality and degree one. Its radical is $\ker q^*$; the Milnor pullback takes that radical to the primitive vanishing cohomology lattice by naturality of the connecting map. These formulas

specify the central ring, including its degenerate middle pairing, without assuming ordinary Poincaré duality for X_0.

7.5. **The spectral coefficient and the allowed parameters.** Proposition 7.1 and Table 1 verify all hypotheses of Theorem 1.1. Taking $D = L$ gives

$$M = \frac{4^2}{5} - 0 = \frac{16}{5}, \qquad \frac{2\pi \sum_{i=1}^{16} (L \cdot C_i)^2}{M} = 10\pi,$$

which proves the spectral assertion of Theorem 1.3. By Proposition 3.2, the vector $(L{\cdot}C_i)_i = \mathbf{1}$ in this calculation is the same vector that generates the kernel of the vanishing-cycle map. The denominator is the polarized primitive norm from the same integral ring. It is not an independent local coefficient fitted to the spectrum.

The family has one exceptional-area direction. Arbitrary independent node parameters are not implied by the number of nodes. Already at first order, evaluation of quintics at the sixteen planar points has rank fifteen: the degree-five Hilbert function of a plane $(4, 4)$ complete intersection is $\binom{7}{2} - 2\binom{3}{2} = 15$. Cayley–Bacharach makes the unique evaluation relation nonzero at every point. The actual pencil selects a compatible vector nonzero at all nodes. This is sufficient for the fixed-ray theorem and does not supply uniformity for arbitrary independent smoothing rates.

8. Tangent sheaves and Hermitian–Einstein limits

The tangent bundles provide compatible nontrivial holomorphic data on the same transition. Their Hermitian–Einstein connections are intrinsic to the Ricci-flat metrics, but their algebraic extensions across the central fibre require a separate calculation. We make that distinction explicit.

8.1. **The flat limit and its reflexive hull.** For the family (48), put

$$\mathcal{A} = T_{\mathbb{P}^4}|_{\mathcal{X}}, \qquad \mathcal{B} = \mathcal{O}_{\mathcal{X}}(5), \qquad \mathcal{K} = \ker(\mathcal{A} \xrightarrow{d_{\mathbb{P}^4}(F_0 + tG)} \mathcal{B}).$$

Thus $\mathcal{K} = T_{\mathcal{X}/\Delta}$, and set $E_0 = \mathcal{K}|_{X_0}$. Write $Q = \bigoplus_{i=1}^{16} \mathbb{C}_{p_i}$.

Proposition 8.1. *The sheaf $\mathcal{K}$ is flat over the disc and restricts to T_{X_t} for $t \neq 0$. There are identities and an exact sequence*

(57)
$$0 \to E_0 \to T_{X_0} \to Q \to 0, \qquad E_0^{**} = T_{X_0}, \qquad q_* T_Y = T_{X_0}, \qquad R^j q_* T_Y = 0 \quad (j > 0).$$

At each node T_{X_0} is reflexive of depth two, whereas E_0 has depth one and is not reflexive. On each exceptional curve,

$$T_Y|_{C_i} = \mathcal{O}(2) \oplus \mathcal{O}(-1)^{\oplus 2}. \tag{58}$$

Proof. Let $\mathcal{I}$ and $\mathcal{Q}$ be the image and cokernel of the relative Jacobian map. Locally the total space is smooth and the projection has equation $\sum z_j^2 = t$. Hence $\mathcal{Q}$ is a length-one skyscraper at each node, annihilated by t, and vanishes elsewhere near the central fibre. Both $\mathcal{I} \subset \mathcal{B}$ and $\mathcal{K} \subset \mathcal{A}$ have no base-parameter torsion, so they are flat over the smooth one-dimensional base. Tensor the sequences

$$0 \to \mathcal{K} \to \mathcal{A} \to \mathcal{I} \to 0, \qquad 0 \to \mathcal{I} \to \mathcal{B} \to \mathcal{Q} \to 0$$

with the central fibre. The first stays left exact, and the second gives $\ker(\mathcal{I}_0 \to \mathcal{B}_0) = \operatorname{Tor}_1^{\mathcal{O}_\Delta}(\mathcal{Q}, \mathbb{C}) = Q$. The kernel of $\mathcal{A}_0 \to \mathcal{B}_0$ is T_{X_0}, so the snake lemma gives the first sequence in (57).

At a node let $R = \mathbb{C}\{z_1, \dots, z_4\}/(\sum z_j^2)$ and let $\mathfrak{m}$ be its maximal ideal. The tangent module is the kernel of $R^4 \to \mathfrak{m}$. The sequence $0 \to \mathfrak{m} \to R \to \mathbb{C} \to 0$ gives depth one for $\mathfrak{m}$, and the kernel consequently has depth two. It is torsion-free and locally free off the node, hence reflexive on this normal threefold. The sequence with nonzero length-one quotient then gives depth one for E_0. Both sheaves agree off codimension three, so their reflexive hulls agree.

The exceptional locus in the smooth Y has complex codimension two. Holomorphic sections of its tangent bundle extend across those curves. On the common regular locus this identifies q_*T_Y with the reflexive extension T_{X_0} [Stacks, Tags 0EBJ and 0AY6]. For the higher direct images, the tangent-normal sequence over C_i splits because $H^1(\mathbb{P}^1, \mathcal{O}(3)) = 0$, giving (58). If I is the exceptional ideal, $I^n/I^{n+1} = \mathcal{O}(n)^{\oplus(n+1)}$. Tensoring with the tangent restriction gives degrees $n+2$ and $n-1$, whose first cohomology vanishes for every $n \geq 0$. Induction through the thickening sequences therefore gives $H^1(T_Y/I^{n+1}T_Y) = 0$. In an incidence chart the node coordinates are the entries of $\left(\begin{smallmatrix} a & b \\ ua & ub \end{smallmatrix}\right)$; the pulled-back node ideal is $(a, b) = I$. These are exactly the thickenings required by formal functions, not merely an equivalent informal neighborhood model. Formal functions and faithful flatness of completion give $R^1q_*T_Y = 0$ [Stacks, Tag 02OC]. The fibres have dimension at most one, so all higher direct images vanish as well. □

The exceptional splitting rules out locally free descent across the contracted point. In particular the resolution direct image is the reflexive hull, not the flat central fibre E_0. Compatibility is through the common regular tangent bundle and the specified coherent extensions. The exterior identifications on the smoothing side are differentiable, not generally biholomorphic. No single flat family containing both the resolution and smoothing branches is asserted.

Every smooth deformation has its intrinsic relative tangent bundle. Thus this bundle choice introduces no additional existence obstruction on smooth base deformations. The singular flatness calculation above uses the actual one-dimensional family with nonzero linear local smoothing terms. It is not a flatness theorem over an arbitrary higher-dimensional singular deformation base. As a consistency check, proper flatness, (57) and the direct images imply the holomorphic Euler-characteristic identity $\chi(Y, T_Y) - \chi(X_t, T_{X_t}) = 16$, distinct from the topological Euler-characteristic difference 32.

8.2. **Actual metrics and controlled local curvature.** On the resolution use the Kähler class $H + \epsilon^2 L$, the restriction of a positive product class, and on the smoothing use H with $t = \epsilon^3$. Denote either branch by Z_ϵ, with its actual Ricci-flat metric g_ϵ.

Proposition 8.2. *The nontrivial tangent bundles T_Y and T_{X_t} are stable for these classes. Their Ricci-flat tangent connections are irreducible zero-slope Hermitian–Einstein connections. Both branches converge to the same metric completion (X_0, d_{g_0}), homeomorphic to X_0, smoothly on fixed regular compact subsets. Near each node, after rescaling by ϵ^{-2}, the metrics and tangent connections converge with all derivatives to the normalized nonflat resolved-conifold or Stenzel model.*

For each node there is a collar radius r, extended over the cap with $r+\epsilon \asymp \epsilon$ there, and fixed constants such that

$$
\begin{aligned}
&|\nabla^j \operatorname{Rm}(g_\epsilon)| \le C_j (r+\epsilon)^{-2-j}, \qquad j \ge 0,\\
&\operatorname{Vol}\{r<\rho\} \le C\rho^6, \qquad \int_{\{r<\rho\}} |F_{T_{Z_\epsilon}}|^2 dV \le C\rho^2, \qquad \epsilon \le \rho \le \rho_0,\\
&d_{GH}((Z_\epsilon, g_\epsilon), (X_0, d_{g_0})) \le C\epsilon^\theta \quad \text{for some } \theta > 0.
\end{aligned} \tag{59}
$$

All tensor norms, derivatives and volumes in this display use the actual metric. In particular the unscaled tangent curvature-energy measure has no atom at a node.

Proof. Existence of the smooth Ricci-flat metrics is Yau's theorem [Yau78]. The holonomy argument following (19) gives full SU(3). Ricci-flatness makes the tangent connection Hermitian–Einstein of zero slope; irreducibility and the Hermitian–Einstein stability implication give stability [Kob87, Theorem 5.8.3]. These bundles are nontrivial since their second Chern pairing with H is 50.

The actual smoothing expansion was verified in Proposition 2.1. On the resolution all local areas are $\int_{C_i}(H+\epsilon^2 L) = \epsilon^2$. Thus the local AC Kähler classes lie in the restriction image of the single global class L, exactly Assumption R of Benabida's Theorem B [Ben26]; his Example 1.5 also treats this split quintic. The face restrictions give the claimed central and rescaled local limits, with all derivatives. For tangent connections these follow by differentiation of the actual metrics and complex structures.

The positive face metrics and the bounded differentiated coefficients in conic frames give uniform cone comparison on annuli and scaled AC comparison on the core. Every physical derivative costs $(r+\epsilon)^{-1}$; two derivatives give curvature. This proves the first line of (59) and core volume $O(\epsilon^6)$. Integrating the annular volume comparison gives the displayed $C\rho^6$ bound. On a dyadic annulus of radius $s \ge \epsilon$, the tangent curvature energy is bounded by $Cs^{-4}s^6 = Cs^2$. Summing to the core, whose energy is $O(\epsilon^2)$, gives $C\rho^2$. The central cone comparison adds one point in the metric completion at each node and identifies its topology with the algebraic cone, so the completion is homeomorphic to X_0.

We give the global distance estimate explicitly. Each filled region $\{r<\rho\}$ has intrinsic diameter at most $C\rho$: radial annular lengths sum to $O(\rho)$, fixed-radius links have diameter $O(\rho)$, and the core has diameter $O(\epsilon)$. Collapsing the sixteen regions to points changes Gromov–Hausdorff distance by at most $C\rho$. Indeed a shortest quotient chain need visit each collapsed point at most once, and each jump can be replaced by a path of its intrinsic diameter. The same applies to the central space. On the common exterior the actual tensor comparison, and its resolution analogue from Theorem B, give relative error $O((\epsilon/\rho)^{a_0})$. The resulting quotient spaces have uniformly bounded diameter and are bilipschitz with factor $1+O((\epsilon/\rho)^{a_0})$. Consequently

$$
d_{GH}(Z_\epsilon, X_0) \le C\big(\rho + (\epsilon/\rho)^{a_0}\big).
$$

Taking $\rho = \epsilon^{a_0/(1+a_0)}$ proves the last line of (59), with a possibly smaller common exponent for the two branches. □

The energy statement allows nonflat rescaled limits and unbounded pointwise curvature. It supplies neither uniform estimates for arbitrary gauge bundles nor a theorem for their HYM flows. Its role here is to identify the actual tangent geometry compatible with the explicitly calculated transition.

8.3. **Proof of the integrated theorem.**

Proof of Theorem 1.3. Proposition 7.1 establishes the genuine projective family, the chosen resolution and the smooth fibres. The complete intersection calculation in Section 7 gives the torsion-free smooth rings, their Chern classes and pairings. The Thom and Mayer–Vietoris calculations there determine the exterior ring and all cohomological attachments. The contraction and Milnor collapse give the central ring and specialization maps by the relative pair sequences, with the naturality described in Section 7.4. This proves part (1).

Use the collar-compatible trivialization of Proposition 2.1 as the exterior identification in the attachment calculation. Its horizontal lift agrees with the prescribed node collars and is defined over the fixed ray. Two such extensions away from the collars are isotopic along that ray, so they induce the same marked cohomology maps. Thus the analytic and topological link generators can be chosen identically. Proposition 3.2 gives (8), since the hyperplane class H is primitive on the smooth quintic and $r(L) = \mathbf{1}$. The freeness of $H^3(U;\mathbb{Z})$ makes its vanishing sublattice primitive. The disjoint vanishing spheres make it isotropic. The calculated factorization $\Lambda_0 \hookrightarrow \Lambda \twoheadrightarrow W$ gives the displayed Picard–Lefschetz block and its Smith form, proving part (2).

For part (3), the same integral class L has $L \cdot C_i = 1$ and cubic data $(H^3, LH^2, L^2H) = (5, 4, 0)$. It determines the exceptional areas s^2 in class $H + s^2L$. On the central regular locus, Proposition 3.1 represents $L - 4H/5$ by the primitive harmonic form of squared norm $16/5$. Identity (27) identifies its sixteen boundary coefficients with the same $r(L) = \mathbf{1}$. The spectral theorem therefore gives 10π, with its stated count, type and error. Proposition 8.2 supplies the actual metric limits on the two branches. Apply its resolution assertion with $\epsilon = s$ and its smoothing assertion with $\epsilon = t^{1/3}$ separately; no ratio of s and t is used.

Finally, Proposition 8.1 identifies the flat central tangent sheaf and its distinct reflexive hull, and Proposition 8.2 proves the tangent stability and actual connection limits. This proves part (4). These statements use the same central regular tangent bundle; they do not identify the flat central sheaf with the resolution direct image, or assert a flat family containing both branches. The restriction, primitive norm and actual collar identifications above provide the compatibility between the topological and analytic assertions. □

9. The eight-node transition and its two small modes

We apply Theorem 1.2 to the compact transition constructed in stability–degeneration Paper 2, [W26e, Appendix A, Proposition A.1]. This uses its geometric realization and exceptional-curve data as premises. The additional deformation argument below supplies a polarized smoothing with the precise first-order and canonical-bundle properties required by the analytic theorem. This family has dual-edge length four at its nontriangular toric faces; it is outside the length-one admissible class of CY VI [W26k, Definition 2.1].

9.1. **A polarized simple smoothing of the earlier example.** Let $B = \mathbb{P}^1 \times \mathbb{P}^1$, $L = \mathcal{O}_B(1, -1)$, and

$$Z = \mathbb{P}_B(\mathcal{O} \oplus L) \times_B \mathbb{P}_B(\mathcal{O} \oplus L). \tag{60}$$

Use the quotient convention for the tautological classes ξ_1, ξ_2, and write f, g for the base hyperplane classes. Let $\bar{Z}$ be the anticanonical model, $Y \subset Z$ a general anticanonical hypersurface chosen simultaneously nondegenerate for its toric polytope, and $q : Y \to X_0 \subset \bar{Z}$ the induced contraction. Set

$$\mathcal{L}_0 = (-K_{\bar{Z}})|_{X_0}, \qquad H = c_1(q^*\mathcal{L}_0) = (4g + 2\xi_1 + 2\xi_2)|_Y. \tag{61}$$

Proposition 9.1. *The threefold X_0 has eight ordinary double points and the displayed map is a smooth simply connected projective small crepant resolution. There is a proper flat projective smoothing over a disc, with trivial relative canonical bundle and a relatively ample line bundle restricting to the exact $\mathcal{L}_0$, such that every local smoothing function has nonzero first derivative. The smooth fibres are simply connected, and the second Betti numbers of Y and X_t are four and two.*

Proof. The cited construction proves that $\bar{Z}$ has two disjoint singular projective lines with transverse ordinary-node singularities. A general anticanonical section meets each in four simple points. Its projective crepant toric refinement restricts to q, and the resulting smooth Y has Hodge pair $(4, 60)$, trivial canonical bundle and $H^1(\mathcal{O}_Y) = H^2(\mathcal{O}_Y) = 0$. Rationality of the nodes transfers both vanishings to X_0. The model $\bar{Z}$ is Gorenstein Fano, so $\mathcal{L}_0$ is ample and Cartier. The canonical-class calculation in the quotient convention gives (61), and Cartier adjunction gives $\omega_{X_0} \simeq \mathcal{O}_{X_0}$.

In the rational divisor basis (f, g, ξ_1, ξ_2), that proposition calculates four exceptional curves of each of the two types

$$\Gamma_f = (1, 0, 0, 0), \qquad \Gamma_g = (0, 1, -1, -1). \tag{62}$$

Its full rational relation space is given by separate sum-zero conditions on these two groups of four. In particular $(1, 1, 1, -3; 1, 1, 1, -3)$ has full support. The first-order criterion of [RT09, Theorem 1.2 and Section 3, (3.2)] gives a deformation tangent nonzero in every local node-smoothing space. Theorem 1.6 of [Fri26] makes this tangent unobstructed: X_0 is a compact Gorenstein Calabi–Yau threefold, its isolated rational lci singularities are Du Bois, its projective resolution satisfies the $\partial\bar{\partial}$ lemma, and $H^1(\mathcal{O}_{X_0}) = 0$.

We verify projective effectivity and the exact polarization. Choose N so that $\mathcal{L}_0^N$ is very ample and $H^1(\mathcal{L}_0^N) = 0$. Over a small Artin extension, the vanishing of $H^2(\mathcal{O}_{X_0})$ lifts the embedding line bundle, and $H^1(\mathcal{L}_0^N) = 0$ lifts its coordinate sections. The lifted morphism remains a closed immersion over the nilpotent thickening. Thus embedded deformations map formally smoothly to abstract deformations; equivalently the restricted Euler sequence gives $H^1(T_{\mathbb{P}^M}|_{X_0}) = 0$. The corresponding Hilbert-scheme germ is smooth. Retain a lift of the selected tangent and shrink to a neighbourhood S with integral fibres.

Perform the Picard construction over this scheme S before choosing the analytic disc. After an étale neighbourhood, the universal family has a section through a smooth point of X_0 [Stacks, Tag 055S]. Its proper integral fibres give the universal identity $\mathcal{O}_T \simeq p_{T*}\mathcal{O}$ under base change, where p is the universal family projection. The relative Picard scheme exists, and the section supplies a universal Poincaré line bundle, normalized along that section [Kle05, Theorems 2.5(2), 4.8; Exercises 3.11, 4.3 and their Appendix A solutions]. Since $H^2(\mathcal{O}_{X_0}) = 0$, the Picard morphism is smooth at $[\mathcal{L}_0]$ [Kle05, Proposition 5.19]. A further étale neighbourhood gives a section through this point; pulling back the Poincaré bundle extends precisely $\mathcal{L}_0$. Ampleness is open. Both base changes have isomorphic tangent maps at the chosen complex points. A holomorphic disc in the resulting smooth base with the prescribed tangent therefore gives a relatively polarized projective family and retains every nonzero local derivative. No ramified reparametrization or replacement of H by NH occurs.

At a node the local equation is $\sum z_j^2 = \tau_i(t)$ with $\tau_i'(0) \neq 0$, so the total space is smooth. Properness allows the disc to be shrunk so that the entire total space is smooth and the family is Gorenstein. Its relative dualizing line bundle restricts to $\omega_{X_0} \simeq \mathcal{O}_{X_0}$. Since $H^1(\omega_{X_0}) = 0$, cohomology and base change lifts a central nowhere-zero generator to a relative section. Its

zero set has proper image missing the origin. Shrinking the disc removes that image and trivializes the actual relative canonical bundle.

For simple connectivity, apply [BK06, Corollary 1.9] to the nondegenerate toric hypersurface and the smooth projective crepant subdivision in the construction. Its lattice quotient is zero: the relevant dual-polytope vertices include the lattice basis e_1, e_2, f_0, g_0. Thus $\pi_1(Y) = 0$. The common exterior has simply connected links $S^2 \times S^3$, and the two kinds of filling retract onto S^2 and S^3. Van Kampen gives $\pi_1(X_t) = \pi_1(Y) = 0$. Finally the two independent exceptional classes give $b_2(X_t) = 4 - 2 = 2$, as also computed in the earlier proposition. □

These properties verify the general smoothability hypothesis of Hein–Sun and the smoothing hypotheses of Benabida used in Section 2. Locally write $\tau_i(t) = tu_i(t)$ with $u_i(0) \neq 0$; a holomorphic square root of u_i changes the coordinates to the model $\sum z_j^2 = t$ over the same base. The Stenzel metric supplies the exact affine Kähler model. The finite-node construction has fixed positive leading scale ratios, and neither metric theorem imposes $b_2(X_t) = 1$. In particular the actual metrics in class H satisfy the comparisons used in Theorem 1.2.

9.2. **The quotient mass and the two coefficients.** The earlier construction calculates

$$\left(\int_Y D_iD_jH\right)_{i,j} = \begin{pmatrix} 0 & 8 & 4 & 4 \\ 8 & 0 & 12 & 12 \\ 4 & 12 & -8 & 0 \\ 4 & 12 & 0 & -8 \end{pmatrix}, \qquad (D_i) = (f, g, \xi_1, \xi_2). \tag{63}$$

It follows directly from $f^2 = g^2 = 0$, $\xi_i^2 = (f - g)\xi_i$ and $\int_Z fg\xi_1\xi_2 = 1$, by integrating $D_iD_j(-K_Z)^2$; the same calculation gives $H^3 = 320$. The restriction kernel has basis

$$k_1 = g + \xi_1, \qquad k_2 = g + \xi_2, \qquad H = 2k_1 + 2k_2.$$

Choose quotient lifts f, g. By (62), their period vectors have disjoint four-entry unit supports. The energy matrix is $\mathsf{E} = 4I_2$. The intersection blocks are

$$B_{KK} = \begin{pmatrix} 16 & 24 \\ 24 & 16 \end{pmatrix}, \quad B_{DK} = \begin{pmatrix} 12 & 12 \\ 12 & 12 \end{pmatrix}, \quad B_{DD} = \begin{pmatrix} 0 & 8 \\ 8 & 0 \end{pmatrix}.$$

Consequently (42) gives

$$\mathsf{M} = \frac{4}{5}\begin{pmatrix} 9 & -1 \\ -1 & 9 \end{pmatrix}. \tag{64}$$

The orthogonal primitive lifts

$$D_- = f - g, \qquad D_+ = f + g - \frac{3}{5}(k_1 + k_2)$$

have squared masses 16 and 64/5, respectively, and both have squared period norm 8. Thus the generalized eigenvalues of (E, M) are 1/2 and 5/8. Theorem 1.2 proves (10), including the count, eventual simplicity, type and gaps, and completes the proof of Corollary 1.4.

Only rational divisor data are used here. No new integral basis, saturation or full cohomology-ring assertion is made for this family. The application uses the earlier compact construction to derive new information about its actual metric operators. It preserves the distinction established there between tangent-stability walls in the ambient fourfold, toric contraction circuits and Calabi–Yau cycle relations.

10. Three small modes on a four-node toric transition

We now use the original-side nodal partial resolution X_9' constructed in CY VI [W26k, Section 11.1 and Appendix G.1]. That paper supplies its four nodes, a projective small resolution and the integral exceptional-curve relation. The deformation results of CY V supply a prescribed first-order smoothing tangent and its integrability. We specify an exact polarization, verify the remaining family hypotheses, and compute three spectral coefficients from the full quotient harmonic norm. This connects the earlier geometric construction to Theorem 1.2.

The notation distinguishes three spaces. The original X_9 of CY V has two nodes and a degree-seven del Pezzo cone; it does not satisfy our nodal hypotheses. Its partial resolution X_9' has four nodes. The four-node mirror studied elsewhere in CY VI is a different threefold. Only X_9' is used in this section.

10.1. **The earlier nodal model and its polarization.** The eleven rays $v_0, \dots, v_{10}$ of the original-side ambient fan are the columns of

$$\begin{pmatrix} 1 & 0 & -1 & 0 & 0 & -4 & -4 & 3 & 2 & -1 & -2 \\ 0 & 1 & -1 & 0 & 0 & -2 & -2 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & 1 & 1 & 0 & 0 \end{pmatrix}. \tag{65}$$

The first nine are the vertices of the reflexive polytope Δ_9; v_9 is interior to a facet and v_{10} is the pentagon centre. Use the source's integral support vector

$$h = (2,2,2,3,3,3,3,2,2,1,2), \qquad P_h = \{u \in \mathbb{R}^4 : h_i + \langle u, v_i \rangle \geq 0 \text{ for all } i\}. \tag{66}$$

Its normal fan is the source fan Σ' with twenty-one maximal cones. The vertices of P_h are

$$\begin{array}{ccc} (-2,-2,-3,15) & (-2,-2,-3,9) & (-2,-2,15,-3) \\ (-2,-2,9,-3) & (-2,-2,15,15) & (-2,4,-3,3) \\ (-2,4,3,-3) & (-2,4,3,3) & (1,-2,-3,3) \\ (0,-2,-3,3) & (1,-2,-3,1) & (1,-2,3,-3) \\ (0,-2,3,-3) & (1,-2,1,-3) & (1,-2,3,3) \\ (0,2,-3,-1) & (0,2,-1,-3) & (0,2,-1,-1) \\ (1,0,-3,-1) & (1,0,-1,-3) & (1,0,-1,-1). \end{array} \tag{67}$$

Substitution in (66) gives the source cones as the active sets; every omitted inequality has margin at least one. The recession cone is zero: pairing with v_0, v_1, v_2 first forces the first two coordinates to vanish, and the remaining opposite coordinate pairs force the other two to vanish. Thus these integral vertices give a strictly convex Cartier support function. On $P' = X_{\Sigma'}$, the divisor $A = \sum h_i D_i$ is ample and Cartier. Set

$$\mathcal{L}_0 = A|_{X_9'}, \qquad H = c_1(q^*\mathcal{L}_0). \tag{68}$$

Here X_9' is the proper transform of a sufficiently general nondegenerate anticanonical member for Δ_9.

For an explicit smooth ambient refinement, replace the heights by

$$a_i = h_i - \frac{\mathbf{1}_{i=3} + \mathbf{1}_{i=4}}{100} + \frac{i^2}{10^6}. \tag{69}$$

The maximal cones of the normal fan of P_a have ray-index sets

$$\begin{gathered}\{0,1,3,6\},\ \{0,1,3,7\},\ \{0,1,4,5\},\ \{0,1,4,7\},\\ \{0,1,5,6\},\ \{0,2,3,6\},\ \{0,2,3,7\},\ \{0,2,4,5\},\\ \{0,2,4,7\},\ \{0,2,5,6\},\ \{1,3,6,9\},\ \{1,3,7,8\},\\ \{1,3,8,9\},\ \{1,4,5,9\},\ \{1,4,7,8\},\ \{1,4,8,9\},\\ \{1,5,6,9\},\ \{2,3,6,10\},\ \{2,3,7,8\},\ \{2,3,8,10\},\\ \{2,4,5,10\},\ \{2,4,7,8\},\ \{2,4,8,10\},\ \{2,5,6,10\},\\ \{3,6,9,10\},\ \{3,8,9,10\},\ \{4,5,9,10\},\ \{4,8,9,10\},\\ \{5,6,9,10\}.\end{gathered} \tag{70}$$

Each ray matrix has determinant ±1, and each cone lies in a cone of Σ' and in a facet of Δ_9. Consequently its toric fourfold T is smooth and projective and the refinement is crepant. At the four source parallelograms

$$q_1 = (3,6,10,8), \quad q_2 = (2,6,3,7), \quad q_3 = (4,5,10,8), \quad q_4 = (2,5,4,7), \tag{71}$$

the refinement uses the diagonal $\{a,c\}$ of $q=(a,b,c,d)$. Thus the anticanonical proper transform $Y \subset T$ is the source's chosen projective small resolution $q : Y \to X_9'$. Extra refinements over facet-interior strata do not change the hypersurface: those strata map to torus-fixed points missed by a nondegenerate member. In particular the additional nonsimplicial coarse cone with rays $\{2,7,8,10\}$ is contained in a single original facet and contributes no fifth node. These model identifications use [W26k, Proposition 3.4].

10.2. **A first-order family with the prescribed ample class.** We state explicitly the additional deformation step. The nodal specializations of CY V used here formulate the classical criterion; no mixed-cone extension of the spectral theorem is asserted.

Lemma 10.1. *Let Z be a connected normal projective nodal threefold over $\mathbb{C}$ with $\omega_Z \simeq \mathcal{O}_Z$ and $H^1(\mathcal{O}_Z) = H^2(\mathcal{O}_Z) = 0$, admitting a projective small crepant resolution. Suppose the exceptional-curve relation space contains a vector with no zero coordinate. For every ample Cartier line bundle $\mathcal{L}_0$ on Z, there is a proper flat projective smoothing over a disc, with trivial relative canonical bundle and a relatively ample line bundle restricting to exactly $\mathcal{L}_0$, such that every node is smoothed to first order.*

Proof. By [W26j, Theorem 2.5], the full-support relation lifts to a global deformation tangent whose image in every local node-smoothing space is nonzero. The common-volume-form identifications may rescale equation parameters by nonzero factors, which preserve this condition. The deformation base is smooth by [W26j, Theorem 4.5]; its additional degree-six hypothesis is vacuous for nodes.

The projective effectivity construction in Proposition 9.1 applies with this chosen tangent. For completeness, choose N with $\mathcal{L}_0^N$ very ample and $H^1(\mathcal{L}_0^N) = 0$. Across a small Artin extension, the obstruction to lifting this line bundle lies in $H^2(\mathcal{O}_Z)$ and the obstruction to lifting its coordinate sections lies in $H^1(\mathcal{L}_0^N)$. Both vanish. The lifted map remains a closed immersion. Embedded deformations therefore map formally smoothly to abstract deformations, and the Hilbert germ is smooth with a lift of the selected tangent.

Work on a finite-type Hilbert neighbourhood with integral fibres. An étale neighbourhood supplies a section through a smooth point of Z [Stacks, Lemma 37.38.5, Tag 057G]. The relative Picard scheme and its normalized Poincaré bundle then exist as in [Kle05, Theorems 2.5(2), 4.8 and Exercise 4.3]. The Picard morphism is smooth at $[\mathcal{L}_0]$ because

$H^2(\mathcal{O}_Z) = 0$ [Kle05, Proposition 5.19]. A further étale-local section through that point extends $\mathcal{L}_0$ itself. Ampleness persists after shrinking. Both changes of base induce tangent isomorphisms. A disc with the chosen tangent in the resulting smooth base retains every nonzero local derivative. The auxiliary power N neither changes the final polarization nor reparametrizes the disc.

The equations $\sum z_j^2 = \tau_i(t)$ with $\tau_i'(0) \neq 0$ make the total space smooth at the nodes. Properness and smoothness away from them permit shrinking so that the total space and every noncentral fibre are smooth. The relative dualizing line bundle restricts to ω_Z. Since $H^1(\omega_Z) = 0$, cohomology and base change lifts a central nowhere-zero generator to a relative section. Its zero locus has proper image missing the origin; after shrinking it is empty. This trivializes the relative canonical bundle. □

Proposition 10.2. *The model X_9', resolution Y and exact polarization (68) admit a smoothing satisfying all geometric hypotheses of Theorem 1.2. Their second Betti numbers are $b_2(Y) = 6$ and $b_2(X_t) = 3$.*

Proof. The earlier construction gives exactly four nodes and no other singularities. Adjunction and crepancy give trivial dualizing sheaves. On the smooth projective toric fourfold T, the anticanonical sequence $0 \to K_T \to \mathcal{O}_T \to \mathcal{O}_Y \to 0$, toric structure-sheaf vanishing and Serre duality give $H^1(\mathcal{O}_Y) = H^2(\mathcal{O}_Y) = 0$. Rationality of the nodes transfers the vanishings to X_9'. The source's full integral curve-relation lattice is

$$\left\{ \ell \in \mathbb{Z}^4 : \sum_i \ell_i [C_i] = 0 \right\} = \mathbb{Z}(1, -1, -1, 1) \tag{72}$$

by [W26k, Proposition 3.5 and Section 11.1]. Lemma 10.1 therefore supplies the exact polarized first-order smoothing and trivial relative canonical bundle.

For simple connectivity, apply [BK06, Corollary 1.9] to the nondegenerate hypersurface and the smooth projective crepant fan above. The relevant lattice contains all original vertices, in particular the basis v_0, v_1, v_3, v_4 of $\mathbb{Z}^4$. Its quotient, hence $\pi_1(Y)$, is zero. Van Kampen across the simply connected links and fillings gives $\pi_1(X_t) = 0$.

The source divisors span $H^2(Y; \mathbb{Q})$ [W26k, Lemma 3.3]. There are four independent character relations, and $D_9|_Y = 0$ because the facet-interior divisor misses the hypersurface. Thus $b_2(Y) \leq 6$. The nonsingular intersection matrix (74) below proves that six indicated classes are independent. The exceptional span has dimension three by (72), so the attachment sequence gives $b_2(X_t) = 6 - 3 = 3$. □

This constructs a suitable smoothing of the exact earlier central model. CY VI's Theorem 9.25 supplies a proper analytic smoothing with projective fibres; it does not by itself assert the precise relative polarization or first-order condition used here. We do not identify the chosen disc with that theorem's arbitrary smoothing arc, or its metrics with those of a birational mirror model.

10.3. **The three-dimensional harmonic quotient.** Use the rational basis of integral divisor classes

$$(D_2, D_5, D_6, D_7, D_8, D_{10}).$$

No assertion that this is an integral lattice basis is needed. The character relations, $D_9|_Y = 0$ and (66) give

$$H = (6, 18, 18, -12, -10, 8), \qquad H^3 = 8712. \tag{73}$$

The polarized intersection matrix is

$$B = \left(\int_Y D_i D_j H \right) = \begin{pmatrix} -18 & 6 & 6 & 6 & 0 & 0 \\ 6 & 0 & 14 & 0 & 0 & 2 \\ 6 & 14 & 0 & 0 & 0 & 2 \\ 6 & 0 & 0 & -22 & 12 & 0 \\ 0 & 0 & 0 & 12 & -16 & 2 \\ 0 & 2 & 2 & 0 & 2 & -6 \end{pmatrix}, \qquad \det B = -5064192. \tag{74}$$

These are intersection products in the toric ring with character relations $\sum_i \langle m, v_i \rangle D_i = 0$ and the Stanley–Reisner relations of (70). Every maximal cone product integrates to one, and $[Y] = \sum_i D_i$. These relations, together with the displayed rays and cones, determine every entry of (74) and the volume in (73).

For $q = (a, b, c, d)$ with the ac diagonal, the integral source circuit is $e_b + e_d - e_a - e_c$. In the chosen basis the four circuits give

$$C = \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & -1 \\ -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ -1 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}. \tag{75}$$

It has rank three, with the full surviving kernel spanned by

$$k_1 = D_2 + D_7, \quad k_2 = D_5 + D_6 - D_7 + D_{10}, \quad k_3 = D_8 + D_{10}. \tag{76}$$

In particular $H = 6k_1 + 18k_2 - 10k_3$. Choose quotient lifts (D_2, D_5, D_6). The energy and the full Schur complement are

$$E = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}, \qquad M = \frac{1}{15} \begin{pmatrix} 177 & -79 & -79 \\ -79 & 158 & -52 \\ -79 & -52 & 158 \end{pmatrix}. \tag{77}$$

The leading principal minors of E are $2, 3, 4$; those of M are $59/5, 869/9, 17584/45$. Both forms are positive definite. Direct expansion gives

$$\frac{\det(E - \mu M)}{-\det M} = \left(\mu - \frac{1}{7} \right) \left(\mu^2 - \frac{375}{628} \mu + \frac{45}{628} \right). \tag{78}$$

The ordered roots are $1/7$ and $(375 \mp 3\sqrt{3065})/1256$. They are positive and distinct. Proposition 10.2 verifies the geometric assumptions, and Theorem 1.2 now proves Corollary 1.5, including its multiplicities, type and gaps.

The earlier vector $(1, -1, -1, 1)$ is the kernel of the curve-class map A and provides a full-support smoothing tangent. Its annihilator is the three-dimensional period image of $C = A^*$ and counts the small modes. The new coefficients additionally use the global form (74) and projection off all three classes in (76). The mirror equality of relation lattices in CY VI is not used to calculate that norm. Thus the earlier construction and deformation theorems are actual premises of this application, while the quantitative spectral conclusion follows from the analysis of the present paper.

11. Reconstruction and further directions

Theorems 1.1 and 1.3 distinguish the information needed for integral reconstruction from that needed for a leading spectral coefficient. The split-quintic construction determines its global exterior and cohomological attachments. These recover the displayed integral rings, specialization maps and vanishing lattice. For the spectral law, the surviving polarization,

the restriction of the lost divisor class and its primitive intersection norm suffice once the genuine geometric family satisfies the analytic hypotheses. The common restriction map identifies which global classes fail to extend over the smoothing and how they contribute to its small spectrum. This is the specific connection between topology and geometry proved here. It does not invert the spectrum to recover the whole smooth space.

The K3 reconstruction results [W26d, W26c] separate local models from the global complement and its attachment information. The present example makes that distinction outside the K3 bundle-moduli setting. An extension to broader classes of threefolds would require a method for calculating the global data from their algebraic degenerations, rather than leaving an unknown complement ring among the conclusions. The objects reconstructed here are the compact resolved, singular and smoothed threefolds at the level of integral cohomology and the specified maps; the cohomology ring of the complex four-dimensional total family is a separate question.

Theorem 1.2 determines the complete degree-two cluster for any fixed number of lost directions. Its mass removes all surviving harmonic classes, and its lower bound applies on every limiting spectral subspace. Corollary 1.4 uses the compact construction and exceptional data of [W26e, Appendix A, Proposition A.1] to obtain two distinct coefficients on that earlier family. This is a direct geometric dependency within the programme. The relation to the K3 attachment results remains methodological: their local models and specialized reconstruction theorems are not premises of the conifold spectral proof.

Corollary 1.5 supplies a direct connection to the separate Calabi–Yau smoothing programme. CY VI's original-side construction provides X_9', its projective small resolution and four exceptional circuits. The nodal specialization of CY V provides a full-support deformation tangent and smooth deformation base. After extending the specified ample class, the global intersections and the full surviving kernel determine three explicit coefficients for the actual Ricci-flat spectrum. This application uses earlier theorems as geometric premises. The mixed-cone criterion and the mirror relation-lattice theorem have different scopes; neither is silently reinterpreted as a nodal spectral result. The curve-class map and its divisor-evaluation transpose explain the common structure, while the quotient harmonic norm is additional global information.

The finite-rank theorem gives leading limits for each fixed family. It does not give relative errors uniform as positive generalized eigenvalues approach zero in a varying family, nor does its full-kernel projection provide the sharper one-sided remainder proved in rank one.

The present simple smoothing family has fixed nonzero leading node-scale ratios. Allowing different vanishing orders or ratios approaching zero requires actual-metric estimates uniform on that enlarged parameter domain. The separate fixed-cap formula (30) does not supply those global estimates. On the split-quintic resolution, equality of all exceptional curve classes also prevents independent exceptional areas.

The small two-form mode has an exact three-form partner under d, and Hodge duality supplies related modes in degrees three and four. The arguments above do not exhaust the middle-degree small spectrum. Likewise, the iterated limits in the lower bound give a leading asymptotic rather than a matching second-order remainder.

Finally, the tangent construction is intrinsic to the underlying Ricci-flat geometry. A separately specified bundle requires its own coupled deformation, stability and local connection analysis. Its endomorphism Laplacian and HYM evolution are different operators and problems from the real Hodge two-form theorem proved here. The local Hermitian–Einstein and

confined-flow theorems [W26f, W26g] retain a fixed smooth background. Applying their conclusions uniformly through a conifold transition would require new control of the background-dependent inverses, bundle deformation and possible curvature concentration. The no-atom tangent curvature estimate in Proposition 8.2 does not provide this analysis for arbitrary bundles. Independently separating node scales, uniform control through further degeneration of the spectral matrix and general gauge-bundle limits are therefore distinct further steps in the reconstruction programme.

References

[AT15] C. Anné and J. Takahashi, *Partial collapsing and the spectrum of the Hodge–de Rham operator*, Anal. PDE **8** (2015), no. 5, 1025–1050. doi:10.2140/apde.2015.8.1025.

[BK06] V. Batyrev and M. Kreuzer, *Integral cohomology and mirror symmetry for Calabi–Yau 3-folds*, in *Mirror Symmetry V*, AMS/IP Studies in Advanced Mathematics, vol. 38, American Mathematical Society, Providence, RI, 2006, 255–270. doi:10.1090/amsip/038/11. Theorem locators refer to arXiv:math/0505432v1.

[BT82] E. Bedford and B. A. Taylor, *A new capacity for plurisubharmonic functions*, Acta Math. **149** (1982), 1–40. doi:10.1007/BF02392348.

[Ben26] A. O. Benabida, *Asymptotics for resolutions and smoothings of Calabi–Yau conifolds*, Comm. Math. Phys. **407** (2026), article 71. doi:10.1007/s00220-026-05578-5. Theorem and proposition locators refer to arXiv:2503.16702v3.

[CO90] P. Candelas and X. C. de la Ossa, *Comments on conifolds*, Nuclear Phys. B **342** (1990), no. 1, 246–268. doi:10.1016/0550-3213(90)90577-Z.

[CPY24] T. C. Collins, S. Picard and S.-T. Yau, *Stability of the tangent bundle through conifold transitions*, Comm. Pure Appl. Math. **77** (2024), no. 1, 284–371. doi:10.1002/cpa.22135.

[EGZ09] P. Eyssidieux, V. Guedj and A. Zeriahi, *Singular Kähler–Einstein metrics*, J. Amer. Math. Soc. **22** (2009), no. 3, 607–639. doi:10.1090/S0894-0347-09-00629-8.

[Fri26] R. Friedman, *Unobstructed deformations for singular Calabi–Yau varieties*, preprint, February 13, 2026. arXiv:2506.09857v3.

[GMS95] B. R. Greene, D. R. Morrison and A. Strominger, *Black hole condensation and the unification of string vacua*, Nuclear Phys. B **451** (1995), nos. 1–2, 109–120. doi:10.1016/0550-3213(95)00371-X.

[GS15] C. Guillarmou and D. A. Sher, *Low energy resolvent for the Hodge Laplacian: applications to Riesz transform, Sobolev estimates, and analytic torsion*, Int. Math. Res. Not. IMRN **2015**, no. 15, 6136–6210. doi:10.1093/imrn/rnu119.

[HHM04] T. Hausel, E. Hunsicker and R. Mazzeo, *Hodge cohomology of gravitational instantons*, Duke Math. J. **122** (2004), no. 3, 485–548. doi:10.1215/S0012-7094-04-12233-X.

[HS17] H.-J. Hein and S. Sun, *Calabi–Yau manifolds with isolated conical singularities*, Publ. Math. Inst. Hautes Ètudes Sci. **126** (2017), 73–130. doi:10.1007/s10240-017-0092-1.

[Kle05] S. L. Kleiman, *The Picard scheme*, in *Fundamental Algebraic Geometry: Grothendieck's FGA Explained*, Mathematical Surveys and Monographs, vol. 123, American Mathematical Society, Providence, RI, 2005, 235–321. Theorem and exercise locators refer to arXiv:math/0504020v1.

[Kob87] S. Kobayashi, *Differential Geometry of Complex Vector Bundles*, Publications of the Mathematical Society of Japan, vol. 15, Kanô Memorial Lectures, vol. 5, Iwanami Shoten, Tokyo, and Princeton University Press, Princeton, NJ, 1987. Official MSJ re-typesetting, 2013.

[LLW18] Y.-P. Lee, H.-W. Lin and C.-L. Wang, *Towards A+B theory in conifold transitions for Calabi–Yau threefolds*, J. Differential Geom. **110** (2018), no. 3, 495–541. doi:10.4310/jdg/1542423628.

[RT09] S. Rollenske and R. Thomas, *Smoothing nodal Calabi–Yau n-folds*, J. Topol. **2** (2009), no. 2, 405–421. doi:10.1112/jtopol/jtp016. Locators refer to the corrected arXiv:0810.0141v3.

[Ros06] M. Rossi, *Geometric transitions*, J. Geom. Phys. **56** (2006), no. 9, 1940–1983. doi:10.1016/j.geomphys.2005.09.005.

[Row08] J. M. Rowlett, *Spectral geometry and asymptotically conic convergence*, Comm. Anal. Geom. **16** (2008), no. 4, 735–798. arXiv:math/0701383.

[Stacks] The Stacks Project Authors, *The Stacks Project*, `https://stacks.math.columbia.edu`, accessed November 9, 2026. Tags 02OC, 0EBJ, 0AY6, 055S and 057G.

[Ste93] M. B. Stenzel, *Ricci-flat metrics on the complexification of a compact rank one symmetric space*, Manuscripta Math. **80** (1993), 151–163. doi:10.1007/BF03026543.

[W26a] B. J. Wuebben, *Canonical metrics on singular moduli of K3 bundles*, preprint, November 8, 2026. doi:10.5281/zenodo.24060228.

[W26b] B. J. Wuebben, *Exceptional cycles and harmonic metrics in K3 bundle moduli*, preprint, November 7, 2026. doi:10.5281/zenodo.24047888.

[W26c] B. J. Wuebben, *Integral attachments and wall correspondences for K3 bundle moduli*, preprint, November 8, 2026. doi:10.5281/zenodo.24054004.

[W26d] B. J. Wuebben, *Integral reconstruction and quantitative metric degeneration of K3 bundle moduli*, preprint, November 7, 2026. doi:10.5281/zenodo.24047889.

[W26e] B. J. Wuebben, *Nonsimplicial local stability chambers of toric tangent bundles*, preprint, October 29, 2026. doi:10.5281/zenodo.23774770.

[W26f] B. J. Wuebben, *Quantitative local models for Hermitian–Einstein metrics*, preprint, November 7, 2026. doi:10.5281/zenodo.24029090.

[W26g] B. J. Wuebben, *Uniform asymptotic phase for Hermitian–Yang–Mills flow near a polystable bundle*, preprint, November 7, 2026. doi:10.5281/zenodo.24038504.

[W26h] B. J. Wuebben, *Uniform Hermitian–Einstein spectra for bipartite two-step extensions*, preprint, October 27, 2026. doi:10.5281/zenodo.23717660.

[W26i] B. J. Wuebben, *A vanishing-cycle obstruction to smoothing Calabi–Yau threefolds*, preprint, October 6, 2026. doi:10.48550/arXiv.2610.35911.

[W26j] B. J. Wuebben, *Smoothing Calabi–Yau threefolds with nodes and del Pezzo cone points*, preprint, October 7, 2026. doi:10.48550/arXiv.2610.35915.

[W26k] B. J. Wuebben, *Mirror symmetry for conifold relations in Calabi–Yau threefolds*, preprint, November 2, 2026. doi:10.5281/zenodo.23904357.

[Yau78] S.-T. Yau, *On the Ricci curvature of a compact Kähler manifold and the complex Monge–Ampère equation, I*, Comm. Pure Appl. Math. **31** (1978), no. 3, 339–411. doi:10.1002/cpa.3160310304.